Cerebellar perturbation impairs human working memory and degrades spatial tuning throughout cortex

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Abstract
Working memory, the transient maintenance and manipulation of information, is fundamental to human cognition. While working memory is typically thought to rely on frontoparietal cortex, recent neuroimaging evidence suggests the involvement of the cerebellum in a host of cognitive functions, including memory. It is currently unknown whether cerebellar processing is necessary for the persistent maintenance of visual input or if cerebellar signatures of working memory are simply a downstream reflection of cortical activity. Using a combination of functional magnetic resonance imaging (fMRI) and transcranial magnetic stimulation (TMS) in humans, we show that cerebellar perturbation broadly degrades cortical spatial tuning and impairs spatial working memory recall. This impairment matches that observed following perturbation of canonical frontoparietal working memory areas. These findings establish a causal role for the cerebellum in the persistent maintenance of cognitive representations, necessitating a revision of prevailing accounts of human working memory.

Introduction

Working memory (WM) enables the transient maintenance of task-relevant information, supporting flexible, goal-directed behavior.[1] Although extensive prior work has identified a distributed network spanning frontal, parietal, and visual cortices underlying visuospatial WM maintenance,[2–12] the causal mechanisms that sustain mnemonic representations remain incompletely understood.

Traditional models posit that persistent activity in prefrontal and parietal cortices maintains visual representations over short delays.[2,3,13–15] More recent work demonstrates that sensory cortices also contribute to WM maintenance, with stimulus-specific patterns observed during delay periods even in the absence of visual input.[7,16–19] Though not typically considered a key region for working memory, recent evidence indicates the cerebellum actively participates in WM maintenance. A recent fMRI study demonstrated that sub-regions of the cerebellum contain stimulus-specific representations during visual WM. Delay-period activity patterns within lobule VIIb/VIIIa selectively encoded the remembered direction of motion of moving dot patterns, and the correspondence between cerebellar and parietal decoding predicted mnemonic precision.[20] These findings suggest that the cerebellum may participate in the distributed storage of WM content. However, it remains unclear whether this cerebellar activity plays a causal role in sustaining visual WM representations or simply mirrors activity patterns in interconnected cortical areas.[21,22]

The cerebellum's uniform microcircuitry and extensive reciprocal projections with association cortex are thought to implement forward internal models that predict the sensory consequences of actions.[23–27] Such predictive computations may generalize beyond movement to cognition, providing a mechanism for stabilizing internal representations against noise or distraction. Supporting this view, optogenetic studies in rodents have shown that perturbing cerebellar output abolishes selective preparatory ramping in frontal cortex during delayed association tasks that require short-term memory to plan a future directional movement.[28–30] These findings indicate that cerebellar output is critical for the emergence of cortical selectivity. Consistent with this account, recent computational modeling work coupled a recurrent cortical network with a feedforward, cerebellar-like module and found that feedback from this module was critical for sustaining stimulus representations across the delay period in the absence of sensory input signals.[31] However, these animal paradigms exclusively involve the maintenance of a prospective motor plan, making it uncertain whether cerebellar computations causally support the maintenance of non-motor spatial WM representations in humans.

Here we combine transcranial magnetic stimulation (TMS), fMRI, and computational modelling to test the causal contribution of the cerebellum to spatial WM in human participants. By integrating causal perturbation and model-based fMRI, this work moves beyond correlational evidence to delineate the causal computational role of cerebellar circuits in cognition. We find that cerebellar TMS induces WM deficits and a broad degradation of spatial tuning across cortex, mirroring the effects of fronto-parietal stimulation. This work establishes that the cerebellum causally contributes to WM, thus challenging prevailing models of the neural underpinnings of distributed WM maintenance. These findings bolster claims that predictive models implemented in cortico-cerebellar loops support not only action but cognition more broadly.

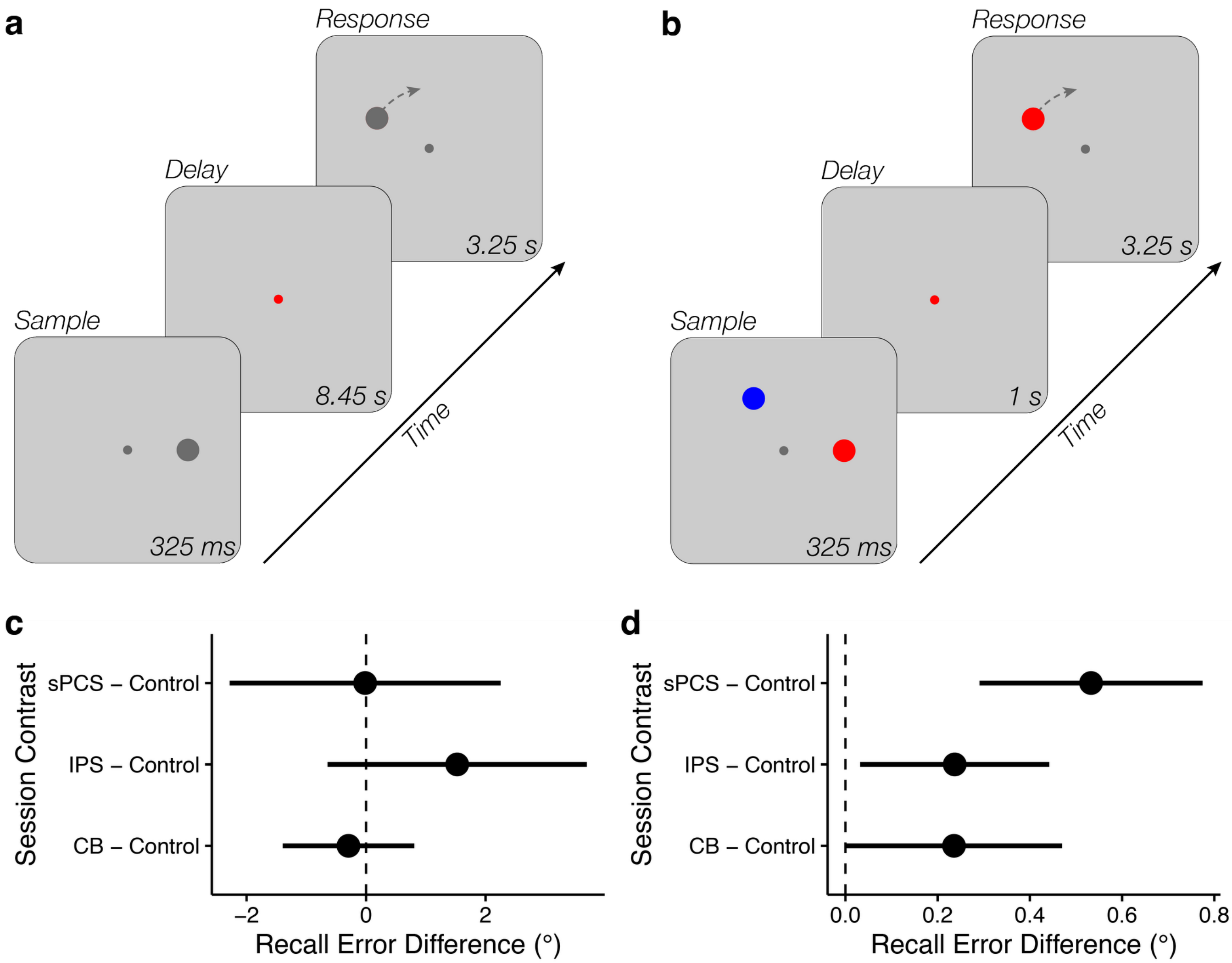


Figure 1. Spatial working memory (WM) paradigms and stimulation induced recall deficits. (a) Inside an fMRI scanner (Experiment 1), participants were presented with a disc stimulus (0.75°) at 4° eccentricity and were instructed to remember the location. 325 ms following the offset of the stimulus presentation period, participants were cued as to whether the trial was an 'active' or 'drop' trial (120 active trials / 20 drop trials). On 'active' trials, participants were instructed to maintain the location of the sample stimulus over the subsequent 8.45 s delay period, while on 'drop' trials they were instructed to drop the stimulus from memory and make a random response. The long delay period allowed us to isolate the BOLD response associated with WM maintenance. Following the delay period, participants were given 3.25 s to adjust a probe disc stimulus to match the remembered location. The initial angle of the probe stimulus was randomized with respect to the memorized location, which ensured that participants were unable to prospectively plan their upcoming motor response (i.e., direction of rotation) during the delay period. Thus, delay-period activity can be inferred to relate to visuospatial WM maintenance rather than preparatory motor activity. (b) In a follow-up psychophysics experiment, participants performed a similar WM task with a larger set size (2 items) and a shorter delay-period (1 s; 480 total trials). A retroactive cue presented 325 ms following the offset of the sample display indicated which of the two stimuli was to be maintained over the delay period. The probe stimulus was again randomized to prevent prospective motor planning. (c,d) Cortico-cerebellar TMS produces WM recall deficits. Parietal stimulation reliably increased recall error in the fMRI experiment (c), while cerebellar, parietal, and frontal stimulation increased recall error in the higher-powered psychophysics experiment (d).

Results

We combined TMS and fMRI to test whether the cerebellum contributes causally to the maintenance of spatial working memory. Participants completed a multi-session protocol. A baseline fMRI session collected population receptive field (pRF) mapping, resting-state scans, and anatomical MRI. The pRF data served two purposes: it allowed us to (1) define individualized TMS targets using functional connectivity and retinotopic constraints, and (2) train a spatial encoding model for subsequent analysis of working-memory representations. Four additional sessions involved the application of continuous theta-burst stimulation (cTBS) over retinotopically organized regions in cerebellar lobule VIIb (CB), intraparietal sulcus (IPS0), and superior precentral sulcus (sPCS), or a somatosensory control site just prior to performing a spatial working memory task in an fMRI scanner. During the TMS-fMRI sessions, participants performed a spatial working memory task in which a single stimulus had to be remembered over a delay period. To improve sensitivity to TMS effects on behavior, we performed a follow-up psychophysics experiment with 4 additional sessions outside the scanner with a shortened delay (1 s vs. 8.45 s), higher set size (2 items vs. 1 item), and substantially increased trial count (480 trials vs 120 trials), thus increasing measurement precision and power to resolve session-dependent differences.

To characterize the effects of TMS on mnemonic precision and underlying neural coding, we combined analyses of behavior, model-based fMRI decoding, and information-theoretic measures of neural representational fidelity. Behavioral performance was evaluated in terms of spatial recall error across stimulation sites. fMRI data were modeled with a spatial encoding framework that predicted voxel-wise BOLD responses from population receptive field parameters derived from the baseline mapping session. This model was then used to (1) decode remembered locations from distributed activation patterns and the uncertainty with which those locations were represented, (2) quantify voxel-level modulation of spatial tuning, and (3) estimate Fisher information to quantify the representational capacity within each region of interest. Together, these analyses tested whether perturbation of the cerebellum, intraparietal sulcus, or frontal cortex altered the information content and fidelity of spatial working-memory representations across the brain.

Cerebellar, parietal, and frontal stimulation impair mnemonic precision
We tested whether perturbing cerebellar or frontoparietal sites altered spatial working-memory precision. Session effects were assessed with hierarchical within-subject permutation tests and FDR correction.

In the TMS-fMRI experiment (Experiment 1; 120 active trials/session), spatial recall was largely unchanged for CB and sPCS (CB – Control = –0.304°, p = 0.76 corrected; sPCS – Control = –0.025°, p = 0.96 corrected), whereas IPS stimulation increased recall error relative to control (IPS – Control = 1.54°, p = 0.005 corrected). This IPS effect remained significant after regressing out session order (p = 0.005 corrected).

In the TMS-psychophysics experiment with higher statistical power (Experiment 2; 480 active trials/session), cerebellar, parietal and frontal stimulation all resulted in increased recall error relative to control stimulation (sPCS – Control = 0.525°, p = 0.0006 corrected; CB – Control = 0.221°, p = 0.008 corrected; IPS – Control = 0.225°, p = 0.008 corrected). The cerebellar and sPCS effects remained significant after controlling for session order (sPCS – Control: p = 0.0006 corrected; CB – Control: p = 0.04 corrected; IPS – Control: p = 0.13 corrected).

Across the two experiments, we observe behavioral deficits associated with each targeted region: IPS in Experiment 1, and cerebellum and sPCS in Experiment 2, where the larger trial count provided greater sensitivity to detect changes in spatial recall precision.

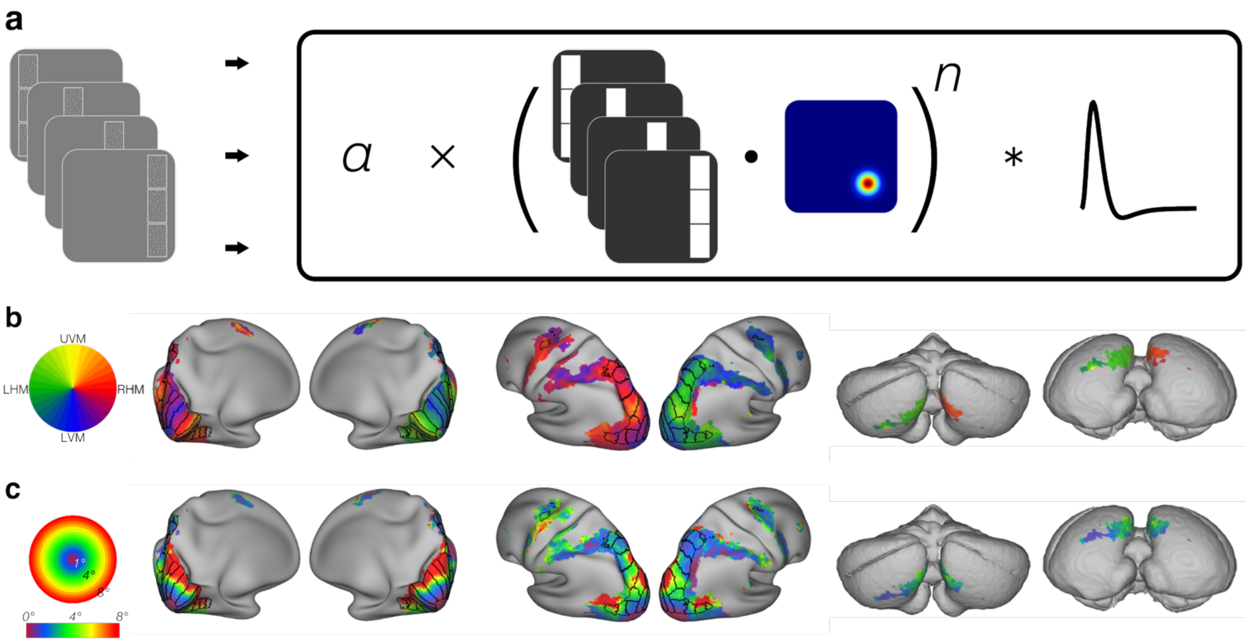


Figure 2. Population receptive fielding (pRF) mapping. (a) In an initial fMRI session, participants performed an attentionally demanding visual field mapping task which required discriminating and comparing the direction of motion of three equal sized patches of moving dot patterns at different positions within the display.[32] Resulting BOLD responses were modeled with compressive spatial summation model.[33] This model produces a 2D Gaussian receptive field for each voxel (x, y, σ) along with gain (a) and compressive nonlinearity (n) parameters. (b) Group average (N = 27) polar angle preferences across cortex and cerebellum. These results replicate prior work demonstrating spatial selectivity in cerebellum.[34,35] (c) Group average eccentricity preferences across cortex and cerebellum.

Cerebellar and frontoparietal stimulation reduce the fidelity of spatial working-memory representations

To assess how perturbing different nodes of the cortico-cerebellar network influenced the fidelity of spatial working-memory representations, we applied a generative probabilistic decoding approach.[20,36–38] This analysis uses voxelwise spatial tuning parameters (Fig. 2) derived from the baseline pRF session to estimate, for each trial, the posterior probability of each possible remembered location (0°–359° at 4° eccentricity) given the observed multivoxel activation pattern during the memory delay (Fig. 3). Averaging trial-aligned posterior distributions produces a session-level probability density over stimulus locations for each region of interest (ROI). Sharper peaks around 0° reflect better decoding of the remembered spatial location. Control stimulation consistently produced more sharply peaked distributions relative to CB, IPS, and sPCS stimulation (see Fig. 4a). From trial-level posterior distributions, we derive two measures: estimation efficiency, the inverse of mean circular squared error between the posterior mean and the true stimulus, and decoded uncertainty, the mean posterior standard deviation across trials.

Cerebellar stimulation produced robust and spatially distributed reductions in the fidelity of spatial representations. Relative to control stimulation, CB stimulation significantly broadened posterior distributions in cerebellar lobule VIIb and a higher-order visual area (V3AB) known to exhibit robust brain-behavior correlations during WM[38] (all $p \leq 0.0004$ corrected; Fig. 4c). This result indicates that cerebellar perturbation increased uncertainty in the representation of remembered locations across both cortical and cerebellar regions. Similar increases in uncertainty were also observed in higher-order parietal regions IPS1 and IPS3 (Supplementary

Fig. 2). In addition to these increases in decoded uncertainty, cerebellar stimulation selectively reduced estimation efficiency in V3AB (p = 0.02 corrected) and IPS0 (p = 0.02 corrected) (Fig. 4b), reflecting a measurable decline in the accuracy with which remembered locations could be reconstructed from population activity. These results indicate that cerebellar perturbation broadly degrades the fidelity of spatial representations and additionally reduces reconstruction accuracy in key visual and parietal regions.

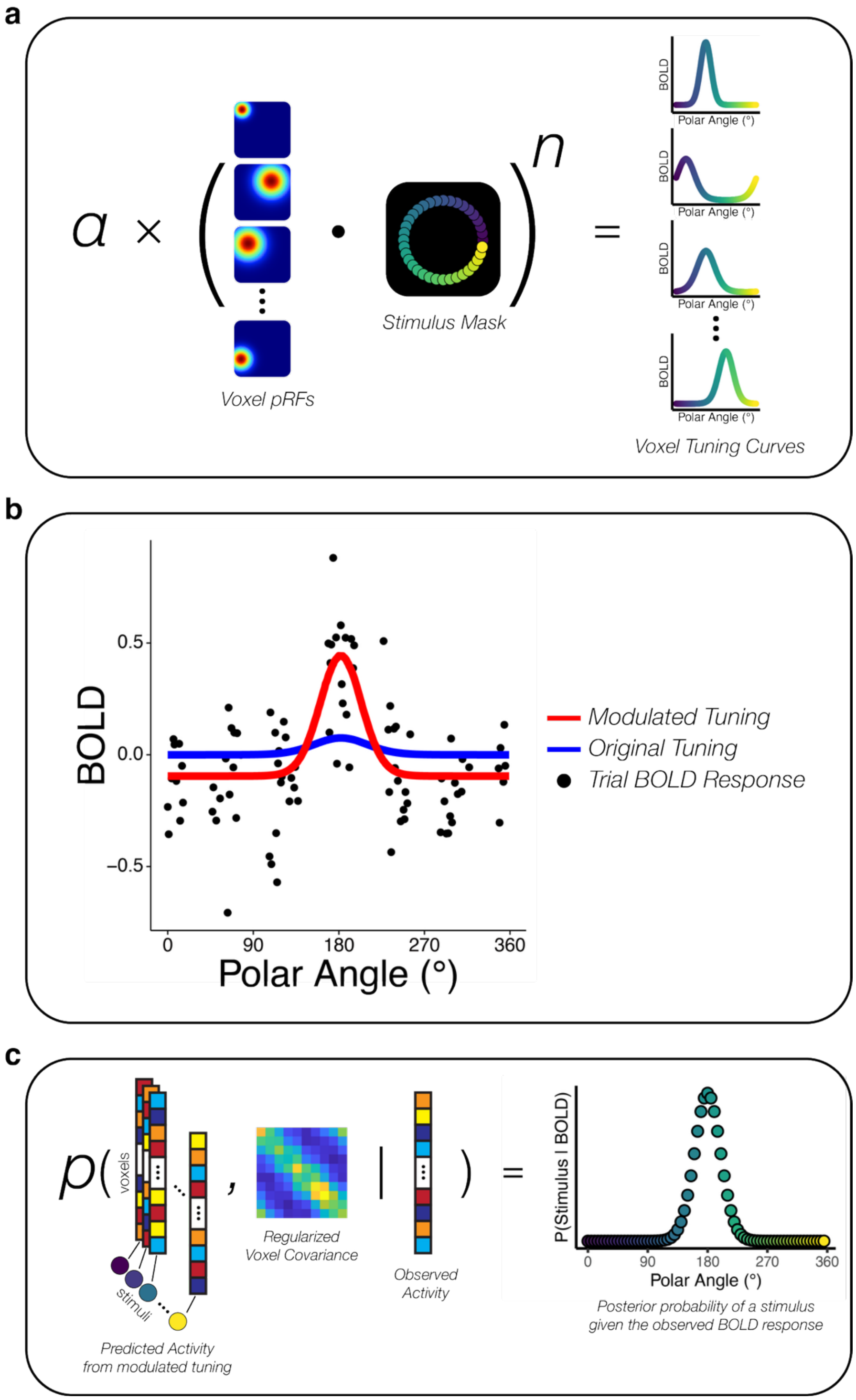


Figure 3. Probabilistic spatial encoding model. (a) fMRI data were modeled with a spatial encoding framework that predicted voxelwise BOLD responses from population receptive field parameters derived from the baseline mapping session (Fig. 2a) and stimulus masks corresponding with all possible spatial WM locations probed in the TMS-fMRI sessions. (b) Spatial tuning curve modulation for an example voxel. The model accounted for session-specific modulation of spatial tuning curves. Each black dot represents the late-delay BOLD response for a single WM trial. The blue curve represents the predicted response to each spatial location from the original model parameters, while the red curve reflects an adjustment in gain, width, and baseline activation as a function of task. (c) Probabilistic decoding of spatial location. For each trial, we computed the posterior probability of each possible stimulus location given the observed multi-voxel BOLD activity on that trial, while taking into account covariance across voxels (see methods).

Parietal stimulation produced a comparable pattern of representational disruption, primarily expressed as increased uncertainty. IPS stimulation significantly increased decoded uncertainty in V3AB, IPS0, and cerebellar lobule VIIb (all $p \leq 0.0004$ corrected; Fig. 4c), as well as in IPS1 and IPS3 (Supplementary Fig. 2). In contrast to cerebellar stimulation, IPS stimulation did not reduce estimation efficiency in any region, indicating that parietal perturbation predominantly affected the precision, rather than the accuracy, of decoded representations.

Frontal stimulation produced a more limited pattern of network-level effects. Relative to control, frontal stimulation increased decoded uncertainty in V3AB and cerebellar lobule VIIb (both $p \leq 0.0004$ corrected; Fig. 4c), indicating reduced spatial selectivity in posterior cortical and cerebellar representations. Effects in IPS0 and other parietal regions were weaker or absent, and estimation efficiency was not significantly altered by sPCS stimulation in any ROI. Thus, unlike cerebellar or parietal perturbation, frontal stimulation exerted more circumscribed effects that were expressed primarily in downstream visual and cerebellar regions rather than broadly across the parietal cortex.

The sPCS region itself exhibited an unusual decoding profile across all stimulation conditions that warrants separate consideration. Under control stimulation, sPCS exhibited weak spatial selectivity, as evidenced by high decoded uncertainty, approaching the values expected under a near-uniform posterior. Following stimulation of all sites (CB, IPS, or sPCS), decoded uncertainty in sPCS decreased relative to control. However, these narrower posterior distributions were not centered on the veridical stimulus location (Fig. 4a), resulting in numerically *worse* estimation efficiency. Thus, reduced uncertainty in sPCS did not reflect enhanced spatial selectivity but rather a qualitative change in the selectivity profile across stimulus locations.

Taken together, the decoding results indicate that perturbation of cerebellar and parietal nodes broadly reduces the spatial selectivity of working memory representations across the cortico-cerebellar network, with cerebellar stimulation additionally reducing reconstruction accuracy in key visual and parietal regions. Frontal stimulation exerts more limited causal effects, primarily increasing decoded uncertainty in posterior cortical and cerebellar regions while leaving estimation efficiency largely unchanged. These findings motivate further examination of how TMS alters the underlying tuning properties of voxel responses.

Cortico-cerebellar TMS alters voxel-level tuning across the cortico-cerebellar network

The probabilistic decoding analysis demonstrated that cerebellar, parietal, and frontal stimulation reduced the precision of spatial working-memory representations across the brain. To identify the tuning mechanisms underlying these effects, we examined how stimulation modulated voxel-level spatial tuning parameters estimated from the generative encoding model. This analysis separately quantified multiplicative gain (ϕ_{gain}), tuning width (ϕ_{size}), additive baseline shifts ($\phi_{baseline}$), and residual variance (ϕ_{var}) relative to the baseline pRF-predicted response. For gain, width, and variance, values greater than 1 indicate increased estimates relative to the tuning estimated in the baseline session, whereas values less than 1 reflect reductions. For the baseline parameter, positive values indicate increases relative to the baseline and negative values reflect decreases. However, as the compressive spatial summation pRF model produces only positive response values, note that negative baseline modulation values do not suggest a true reduction from the baseline session. Rather, they reflect the model's adjustment to z-scored time series with zero mean, as the BOLD response during working memory maintenance has a smaller magnitude than the response to visual stimulation and motor responses.

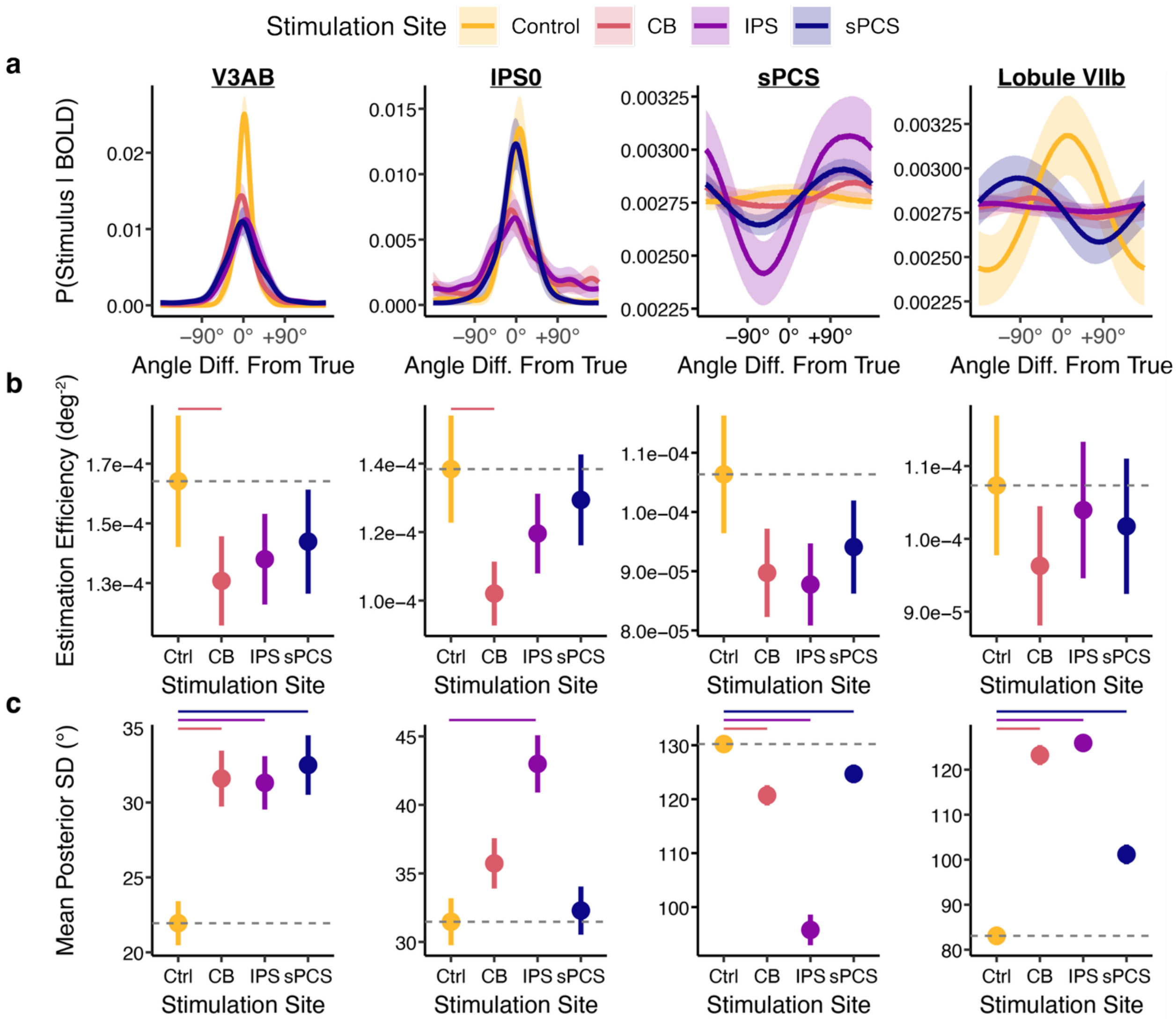


Figure 4. Cortico-cerebellar stimulation degrades spatial decoding across multiple brain regions. (a) Mean posterior probability distribution. Averaging trial-aligned posterior distributions produces a session-level probability density over stimulus locations for each region of interest (ROI). Sharper peaks around 0° reflect better decoding of the remembered spatial location. (b) Estimation efficiency (inverse of mean circular squared error between the posterior mean and the true stimulus) reflecting decoding accuracy for each session and region. (c) Mean posterior standard deviation reflecting decoding uncertainty. Error bars reflect a bootstrap estimate of standard error. Horizontal bars indicate corrected p < 0.05 from a hierarchical permutation test. Each column reflects results for one ROI. IPS – intraparietal sulcus; sPCS – superior pre-central sulcus.

Cerebellar stimulation produced the most widespread and consistent tuning changes across the network. Relative to control stimulation, cerebellar TMS significantly reduced the additive baseline parameter in all primary ROIs, including V3AB, IPS0, and cerebellar lobule VIIb (all $p = 0.0002$ corrected), indicating a broad reduction in stimulus-independent BOLD activity. Because the baseline term captures components of the response that do not vary with the spatial location of the stimulus, these shifts reflect a global suppression of visuocortical activity consistent with prior demonstrations that cTBS reduces neural responsiveness across cortical[39–41] and cerebellar targets.[42–46] In addition to these baseline effects, cerebellar stimulation reduced gain in V3AB, IPS0, and VIIb (all $p \leq 0.003$ corrected), increased residual variance in IPS0, sPCS, and VIIb ($p < 0.01$ corrected), and increased tuning width in VIIb ($p = 0.0007$ corrected). These combined changes are consistent with a reduction in both the overall responsivity and the selectivity of spatial tuning functions. Supplementary analyses revealed parallel patterns in early visual cortex (V1–V3) and higher-order parietal regions (IPS1–IPS3), with widespread gain reductions (all $p < 0.002$ corrected) and baseline decreases (all $p < 0.0004$ corrected) accompanied by region-specific variance changes (IPS1 and IPS3; $p = 0.0003$ corrected) (Supplementary Fig. 3, 4; Supplementary Table 2).

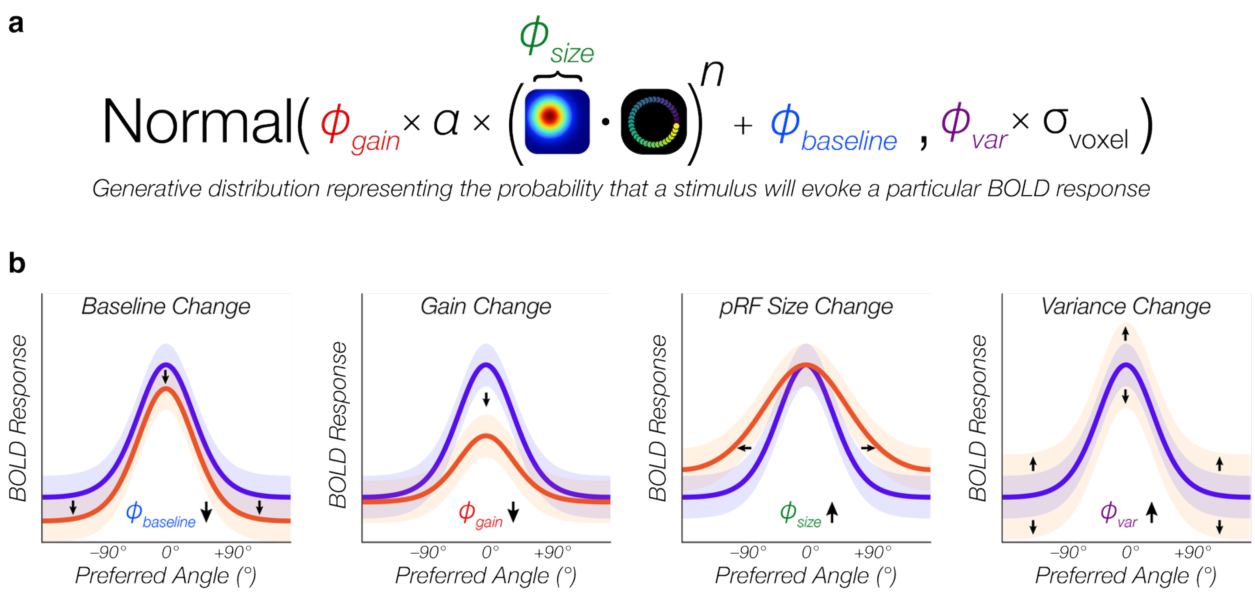


Figure 5. Spatial tuning curve modulation and hypothetical effects of stimulation. (a) Analysis separately quantified TMS-induced changes in multiplicative gain (ϕ_{gain}), tuning width (ϕ_{size}), additive baseline shifts ($\phi_{baseline}$), and residual variance (ϕ_{var}) relative to the baseline pRF-predicted response. Gaussian center (x, y) and compressive nonlinearity exponent (n) were held constant. (b) Possible effects of stimulation on different aspects of voxel spatial tuning.

Parietal stimulation yielded a similar profile of results to cerebellar stimulation. IPS stimulation also produced robust reductions in baseline activity across V3AB, IPS0, sPCS, and VIIb (all $p = 0.0002$ corrected), as well as supplementary ROIs (V1-3 and IPS1-3; all $p = 0.0002$ corrected), consistent with a global suppression of stimulus-independent responses. In contrast to cerebellar stimulation, IPS stimulation exerted more selective effects on tuning parameters governing spatial selectivity. IPS stimulation significantly reduced gain in V3AB, IPS0, and lobule VIIb (all $p \leq 0.0004$ corrected), and increased residual variance in V3AB and VIIb (both $p = 0.0003$ corrected). IPS stimulation decreased residual variance in sPCS ($p = 0.0003$), a pattern that parallels the reduction in decoded uncertainty observed in this region. Across supplementary parietal ROIs, IPS stimulation produced heterogeneous but significant changes in gain, variance, and tuning width, indicating that parietal perturbation alters both global responsivity and region-specific tuning properties across the network (see Supplementary Table 2).

Frontal stimulation produced tuning changes that were directionally similar to cerebellar and parietal stimulation but more spatially circumscribed. Frontal stimulation significantly reduced baseline activity across V3AB, IPS0, and VIIb (all $p = 0.0002$ corrected), and reduced gain in V3AB, IPS1, IPS3, and cerebellar lobule VIIb (all $p \leq 0.003$ corrected). Residual variance increased in V3AB, IPS3, and sPCS itself (all $p \leq 0.002$ corrected). In contrast, effects on IPS0 and IPS2 were weaker or absent. Thus, frontal stimulation reliably altered both stimulus-independent and tuning-related parameters, but these effects were expressed in a narrower subset of regions than those observed following cerebellar or parietal stimulation.

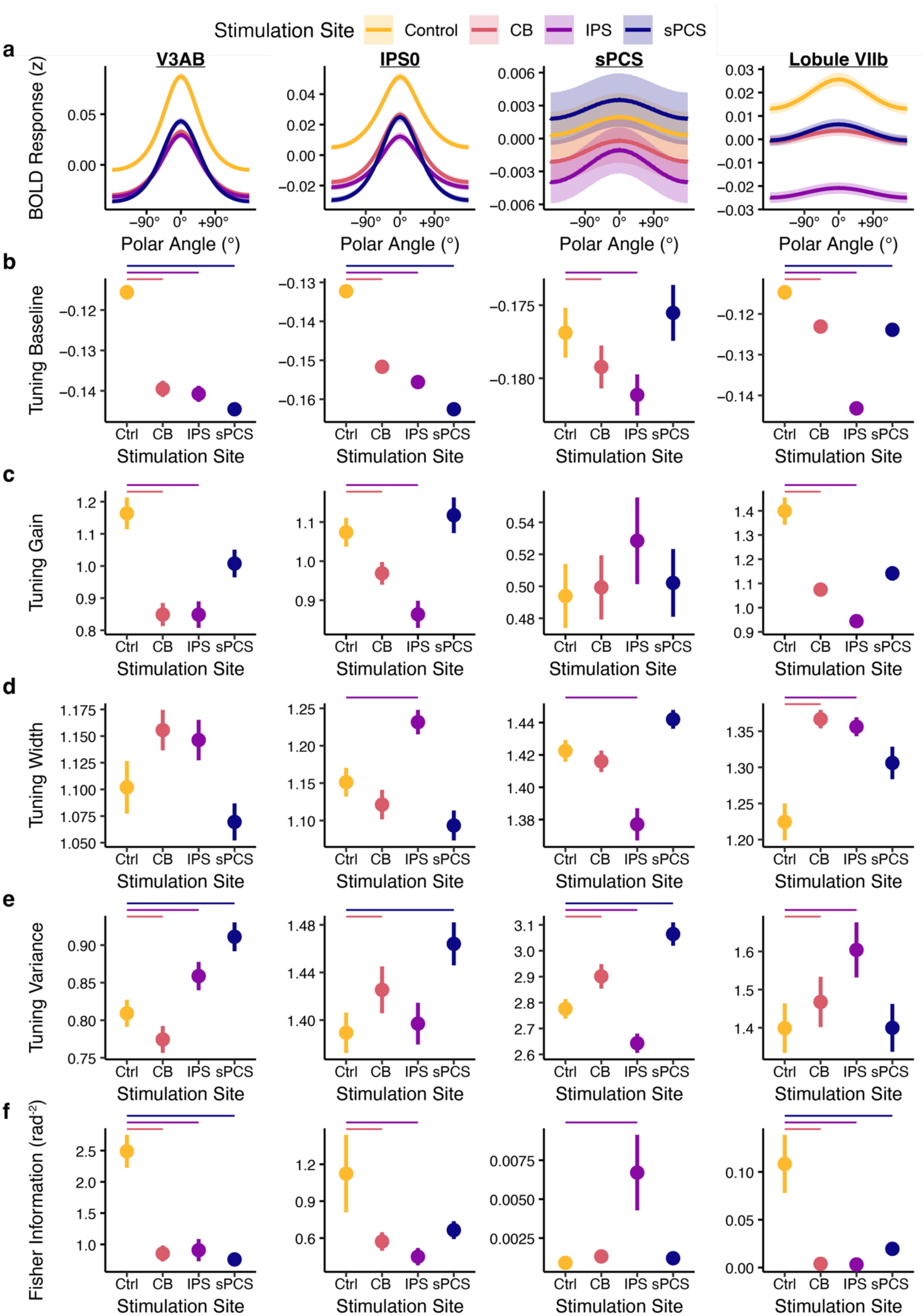
Taken together, cerebellar, parietal, and frontal perturbation produced broadly similar tuning modulation, differing primarily in their spatial extent and regional expression rather than in the qualitative nature of the underlying effects. Reductions in stimulus-independent baseline activity and multiplicative gain were the most consistent effects across stimulation sites, whereas changes in tuning width and residual variance were more region- and stimulation-specific. Although baseline shifts primarily reflect global changes in responsivity, the extent to which population responses vary with spatial location is jointly determined by gain, width, and variance. The observed pattern of tuning modulation therefore raises a key question: to what extent do these changes alter the overall informational capacity of spatial working-memory representations across the cortico-cerebellar network?

Causal disruption of informational capacity across cerebellum and cortex

To assess the net effect of these TMS-induced changes in voxel-level tuning on the representational capacity of the network, we quantified cross-validated Fisher information (FI; $\bar{I}_{cv}$; see methods). This measure reflects how sensitively a population's mean response varies with spatial location relative to its noise covariance. FI therefore characterizes the intrinsic discriminability supported by the underlying population code, independent of trial-by-trial decoder variability. By estimating tuning on training runs and noise covariance on held-out runs, this approach yields an unbiased assessment of representational precision that directly tests whether TMS alters the information-carrying capacity of neural populations across the brain.

Stimulation applied to cerebellum, parietal, and frontal cortex produced significant and spatially distributed reductions in FI. Relative to control, CB stimulation reduced FI in V3AB, cerebellar lobule VIIb, IPS1, and V1 (all $p < 0.006$ corrected). Parietal stimulation significantly reduced FI in V3AB, IPS0, IPS1, IPS3, and cerebellar VIIb (all $p < 0.02$ corrected). Frontal stimulation significantly reduced FI in V3AB, IPS1, IPS3, and cerebellar VIIb (all $p < 0.007$ corrected).

Taken together, these results demonstrate that cerebellar, parietal, and frontal stimulation all causally reduce the informational capacity of neural populations associated with spatial working memory coding. Parietal stimulation produced the broadest loss locally across the parietal hierarchy, whereas cerebellar stimulation produced distributed reductions spanning visual cortex, posterior parietal cortex, and cerebellum. These FI results demonstrate that the net effect of disruption of cerebellum and frontoparietal cortex is not only the altered expressible uncertainty of spatial representations, but also a reduction in the underlying informational capacity of the cortico-cerebellar network during WM maintenance.



Stimulation Site
Control
CB
IPS
sPCS
a
b
c
d
e
f
V3AB
IPS0
sPCS
Lobule VIIb
BOLD Response (z)
Polar Angle (°)
Tuning Baseline
Tuning Gain
Tuning Width
Tuning Variance
Fisher Information (rad⁻²)
Stimulation Site
Ctrl
CB
IPS
sPCS
−90°
0°
+90°

Figure 6. TMS induced changes in spatial tuning during WM maintenance. (a) Mean tuning curve across voxels for each stimulation session within each ROI. For visualization, the tuning curves are demeaned across sessions. Shaded ribbons reflect bootstrap estimate of SEM across voxels. (b) Spatial tuning curve baseline for each session and ROI. This value reflects stimulus-independent BOLD activity during the delay-period. (c) Spatial tuning curve gain for each session and ROI. This value reflects the BOLD response for preferred stimulus locations relative to non-preferred locations. (d) Spatial tuning curve width for each session and ROI. (e) Spatial tuning curve variance for each session and ROI reflecting the distribution of BOLD responses for repeated presentations of the same stimulus. (f) Cross-validated Fisher information for each session and ROI. This value reflects how sensitively a region's mean response varies with spatial location relative to its noise covariance, thus characterizing the intrinsic representational capacity supported by the underlying population code. This measure represents the net impact of all TMS-induced changes in spatial tuning. Error bars across b-f reflect a bootstrap estimate of SEM. Horizontal bars indicate corrected p < 0.05 from a hierarchical permutation test.

Discussion

We provide evidence that the human cerebellum, along with frontoparietal cortex, contributes causally to the maintenance of non-motor spatial working-memory representations by showing that perturbing cerebellar cortex influences coding across the distributed cortico-cerebellar network. We combined individualized cTBS with functional neuroimaging and model-based analyses linking voxel-level tuning to population-level representational readouts. Specifically, we used pRF mapping to define subject-specific targets and to train a spatial encoding model. We then showed that cerebellar, frontal, and parietal stimulation all affected (1) behavioral recall errors, (2) probabilistic decoding of remembered location and trial wise posterior uncertainty during the delay period, and (3) complementary measures of tuning modulation and representational capacity.

By demonstrating that perturbation of frontal and parietal cortices reduces the fidelity of spatial working memory representations, our results are consistent with a large body of work implicating frontoparietal regions in working memory using both correlational and causal approaches.[2,3,5,6,10–13,47,48] Prior work has also implicated the cerebellum in working memory, but the evidence has been predominantly correlational. Neuroimaging and neuropsychological studies have associated cerebellar activity with verbal working memory and articulatory rehearsal processes, particularly in lobules VI and Crus I/II,[49–52] while a separate body of work has linked the posterior cerebellum to visual and spatial working memory demands.[20,34,53–57] Convergent large-scale functional mapping has further indicated that cerebellar areas are systematically coupled with association networks supporting cognitive control, attention, and memory.[58–63] Yet despite this consistency across studies, such findings cannot determine whether cerebellar activity is necessary for working memory maintenance or instead reflects downstream consequences of cortical processing. By combining cerebellar stimulation with model-based fMRI measures of representational fidelity, the present study provides causal evidence that cerebellar output contributes to the maintenance of spatial working-memory representations in humans.

In the current study, behavioral performance was reliably affected by theta-burst stimulation. TMS disruption of cerebellar, parietal, and frontal targets each increased recall error relative to control. Together, these results show that perturbing any node in this cortico-cerebellar circuit can impair spatial recall. Probabilistic decoding analyses of fMRI data revealed that stimulation primarily altered the expressed fidelity of delay-period spatial representations. Across stimulation conditions, delay-period decoded posterior distributions over remembered spatial location were systematically reshaped, indicating changes in the reliability with which population activity specified locations maintained in working memory.

The predominant signature of TMS disruption across the brain was a change in the certainty with which stimulus location could be decoded from BOLD activity. However, broadening of the posterior probability distribution over spatial location does not distinguish whether these effects reflect reduced tuning sensitivity, increased response variability (or covariance), or some combination of both. We therefore complemented the decoding analysis with voxel-level tuning analyses and cross-validated Fisher information to directly evaluate how stimulation affected the signal and noise components that jointly determine the representational capacity of a given brain region. Cortico-cerebellar TMS systematically altered multiple components of voxel tuning. Across stimulation sites, the most consistent effects were reductions in stimulus-independent baseline activity and multiplicative gain, accompanied by more heterogeneous changes in tuning width and residual variance. These tuning modulations indicate that perturbation of cerebellar, parietal, or frontal nodes reshapes multiple components of voxel response profiles across spatially selective regions of the cerebral and cerebellar cortices. This result is consistent with a distributed circuit-level influence of the stimulated sites. Critically, these changes were associated with reliable reductions in cross-validated Fisher information, demonstrating that stimulation degraded the intrinsic information-carrying capacity of the population codes that support spatial working memory. Together with the decoding results, these findings show that network perturbations do not merely alter the trial wise readout of mnemonic representations but modify the representational limits of the system itself, reducing the fidelity with which spatial information can be maintained across the cortico-cerebellar network.

Recent causal evidence from animal models provides a useful framework for interpreting these results. Perturbing cerebellar output can directly reshape cortical dynamics and task-related activity patterns, indicating that cerebellar circuits actively influence forebrain computations rather than merely refining movement execution.[28–30] For example, disrupting cerebellar nuclei abolishes the emergence of selective preparatory dynamics in frontal cortex during a delayed association task,[29] demonstrating that cerebellar output can be necessary for the emergence and maintenance of task-relevant cortical states. More recent work further suggests that cortico-cerebellar interactions are dynamic and state-dependent, varying across behavioral task stages and computational demands.[64] However, because many of these paradigms tightly couple working memory demands with prospective motor planning, it is not possible to ascertain whether cerebellar computations support the maintenance of non-motor representations. Here, by randomizing the initial location of the probe stimulus, we minimize the opportunity for subjects to pre-plan a specific sequence of motor actions during the delay. Under these conditions, cerebellar stimulation reduced the fidelity of delay-period spatial representations, indicating that cerebellar computations influence the maintenance and encoding of remembered spatial information, rather than reflecting motor planning or response selection alone.

How might cerebellar computations support working memory? Prominent theories propose that the cerebellum implements forward internal models that help predict and stabilize neural dynamics.[24–27,31] Consistent with this role, the cerebellum has been extensively implicated in visuomotor adaptation, where it is thought to update internal models based on sensory prediction errors.[23,24,26,65–70] In a working memory context, this view implies that cerebellar output could contribute to maintaining task-relevant cortical states by reducing drift, limiting noise accumulation, or coordinating distributed population codes across frontoparietal and sensory regions. Computational work is consistent with this idea: coupling a recurrent cortical network to a cerebellar-like module improves the stability of stimulus representations across delays without sensory input.[31] Together, these results support the view that cerebellar computations help sustain high-fidelity cortical representations during memory maintenance.

We recently provided behavioral evidence that further supports the idea that cerebellar adaptive control mechanisms traditionally associated with motor learning extend to spatial cognition. In a task that repeatedly induced covert attentional allocation errors, we found that spatial working memory representations gradually shifted to counteract those errors, exhibiting exponential time courses, spatially graded transfer, and rapid de-adaptation.[71] These signatures closely parallel canonical sensorimotor adaptation. The present findings identify a potential neural substrate for such adaptive stabilization, demonstrating that perturbing cerebellar output degrades the fidelity of maintained spatial representations. Future work is needed to determine whether cerebellar contributions to working memory are computationally distinct from those of frontoparietal cortex, perhaps reflecting a specific role in error-based calibration of mnemonic representations.

These results inform ongoing debates regarding the neural substrates of working memory. Sensory recruitment accounts propose that mnemonic content is maintained within stimulus-selective sensory populations,[1,4,7,16,72] whereas alternative views emphasize higher-order or distributed storage mechanisms.[9,73,74] The present data support a distributed perspective: perturbing cerebellar, parietal, and frontal nodes altered representational fidelity, and cerebellar disruption impacted information-bearing capacity in canonical visuospatial regions. Together, these findings challenge cortico-centric accounts of working memory by demonstrating that the persistent maintenance of cognitive representations depends critically on cerebellar contributions to a broader maintenance network.

Methods

Design overview. We used a within-subject, multi-session protocol to test how cerebellar, parietal, and frontal visuospatial maps support spatial working memory (WM). A baseline MRI session provided high-resolution anatomical images, resting-state fMRI, and retinotopic pRF mapping. These data served two roles: (1) defining individualized TMS targets using retinotopy and functional-connectivity constraints within atlas-based anatomical search spaces (cerebellar lobule VIIb (CB), intraparietal sulcus area IPS0 (IPS), superior precentral sulcus (sPCS), and a somatosensory control site); and (2) training a spatial encoding model for subsequent spatial decoding analyses of the spatial WM task.

Each participant then completed four TMS-fMRI sessions, one per stimulation site (CB, IPS, sPCS, control). At the start of each session, continuous theta-burst stimulation (cTBS) was delivered at an individually titrated intensity. Immediately afterward, participants performed the spatial WM task during fMRI while cTBS after-effects were present.[39] Session order was counterbalanced across participants. The primary fMRI outcomes were decoding-based estimates of remembered location and uncertainty, voxel-wise tuning-curve modulation, and cross-validated Fisher information. Each of these measures derived from the fMRI data following stimulating a site of interest was contrasted with that following stimulation of the control site.

A subset of participants completed additional TMS-psychophysics sessions outside the scanner (same four stimulation sites). Experiment 2 used a slightly modified WM task relative to the fMRI sessions in Experiment 1 and was more strongly powered to detect subtle behavioral effects (detailed below).

Participants. 28 total healthy adult volunteers participated in this study. All research protocols were approved by the Health Sciences and Behavioral Sciences Institutional Review Board at the University of Michigan. All participants gave written informed consent. Participants were paid $15/hr for their participation. A $50 bonus payment was disbursed if a participant successfully completed all sessions. Participants were recruited from the University of Michigan and the surrounding community. All participants possessed normal or corrected-to-normal vision.

Due to the longitudinal nature of this within-subject design and the logistics of scheduling up to 9 total sessions, many participants did not complete all possible sessions. 9 participants did not return for the TMS-fMRI sessions, but their baseline sessions were important for training the pRF model to assess the impact of TMS on spatial representation (see below). Of the combined TMS-fMRI sessions, 17 subjects completed the Control session, 17 subjects completed the sPCS session, 16 subjects completed the CB session, and 16 completed the IPS session. 14 of the original participants were invited to complete an additional 4 sessions combining TMS with psychophysics. 12 subjects completed all sessions. Of the two remaining subjects, one completed only the control session and the other completed all sessions except for the CB session.

MRI Acquisition. MR data were acquired with a GE Signa MR750 3.0T MRI scanner with a 32-channel head coil. The baseline session collected high-resolution 3D inversion-recovery T1-weighted MPRAGE (sagittal; TR = 2523 ms; TE = 3.55 ms; TI = 1060 ms; flip angle = 8°; 0.8 mm isotropic voxels; matrix = 320 × 320; 208 slices; parallel imaging: auto-calibrating reconstruction for cartesian imaging (ARC) 2x in-plane and 2x through-plane) and 3D CUBE T2-weighted scans (sagittal; TR = 5202 ms; TE = 60 ms; flip angle = 90°; 0.8 mm isotropic voxels; matrix = 320 × 320, 226 slice; parallel imaging: ARC 2x in-plane and 2x through-plane). Functional data for resting-state and task scans were acquired with a 2D gradient-echo echo-planar imaging (EPI) sequence using simultaneous multi-slice acceleration (GE HyperBand; MB = 8): TR = 650 ms; TE = 35 ms; flip angle = 52°; 2.3 mm isotropic voxels; matrix = 92 x 92; 64 slices with 0% inter-slice gap. Spin-echo EPI field maps were also acquired with opposite phase encoding directions (Anterior-to-Posterior; Posterior-to-Anterior) and geometry matched to the functional EPI timeseries.

MRI Preprocessing. Anatomical, functional, and resting-state data were preprocessed using *fMRIPrep*[75] (Version 20.2.5; RRID:SCR_016216), which is based on *Nipype*[76] (Version 1.6.1; RRID:SCR_002502), and the *ciftify* package[77] (Version 2.3.3). The T1-weighted (T1w) image was corrected for intensity non-uniformity (INU) with *N4BiasFieldCorrection*,[78] distributed with ANTs 2.3.3[79] (RRID:SCR_004757), and used as the T1w-reference throughout the workflow. The T1w-reference was then skull-stripped with a *Nipype* implementation of the *antsBrainExtraction.sh* workflow (from ANTs), using OASIS30ANTs as target template. Brain tissue segmentation of cerebrospinal fluid (CSF), white-matter (WM) and gray-matter (GM) was performed on the brain-extracted T1w using *fast*[80] (FSL 5.0.9; RRID:SCR_002823). Brain surfaces were reconstructed using *recon-all*[81] (FreeSurfer 6.0.1; RRID:SCR_001847), and the brain mask estimated previously was refined with a custom variation of the method to reconcile ANTs-derived and FreeSurfer-derived segmentations of the cortical gray-matter of Mindboggle[82] (RRID:SCR_002438). Freesurfer outputs were then input into *ciftify_recon_all*, which performs surface-based alignment using the MSMSulc algorithm[83] and resamples data from native space to 32k fsLR space. The fsLR 32k space combines the benefits of nonlinear MNI volumetric registration (for subcortical structures; FSL FNIRT) with surface-level registration based on sulcal anatomy for the cortex.[84]

For each BOLD run (across all tasks and sessions), the following preprocessing was performed. First, a reference volume and its skull-stripped version were generated using a custom methodology of *fMRIPrep*. A B0-nonuniformity map (or *fieldmap*) was estimated based on two (or more) echo-planar imaging (EPI) references with opposing phase-encoding directions, with *3dQwarp*[85] (AFNI 20160207). Based on the estimated susceptibility distortion, a corrected EPI (echo-planar imaging) reference was calculated for a more accurate co-registration with the anatomical reference. The BOLD reference was then co-registered to the T1w reference using *bbregister* (FreeSurfer) which implements boundary-based registration.[86] Co-registration was configured with six degrees of freedom. Head-motion parameters with respect to the BOLD reference (transformation matrices, and six corresponding rotation and translation parameters) were estimated before any spatiotemporal filtering using *mcflirt*[87] (FSL 5.0.9). The BOLD time-series were resampled onto their original, native space by applying a single, composite transform to correct for head-motion and susceptibility distortions. Gridded (volumetric) resamplings were performed using *antsApplyTransforms* (ANTs), configured with Lanczos interpolation to minimize the smoothing effects of other kernels.[88] For the definition of TMS target regions, resting-state and pRF timeseries were analyzed in native T1 space. These native T1w timeseries were smoothed in a parcel-constrained manner (cortical ribbon and subcortical gray matter structures) with a 2mm FWHM kernel to maximize spatial accuracy of the TMS target definitions. For resting-state timeseries, this smoothing was done after additional preprocessing was performed (see *resting-state preprocessing* below).

Using the *ciftify_subject_fmri* function, pRF and spatial working memory fMRI acquisitions were then projected to the native surface in a weighted, ribbon-constrained manner, where the value assigned to each vertex is calculated as the weighted average of the voxels encompassed (or partially encompassed) by the cortical ribbon. Noisy voxels, as assessed by their local coefficient of variation, were excluded from this operation. Surface data was then resampled to fsLR 32k surface space. Subcortical data were nonlinearly resampled to MNI152 within their anatomical gray matter masks.[77,84] Surface and volumetric data were then combined into a Connectivity Informatics Technology Initiative (CIFTI) "grayordinate" file format. Lastly, CIFTI smoothing was applied (2D surface smoothing and parcel-constrained smoothing for subcortical volumes) using a 4mm FWHM kernel. Prior to any analysis, the pRF and spatial working memory run timeseries of each voxel/vertex were detrended with a 2nd-order polynomial, converted to percentage signal change, and then normalized (z-score) across time points within each run.

Resting-state preprocessing. Following fMRIPrep, we additionally performed band-pass filtering and confound regression on resting-state timeseries using the *nilearn* python package (Version 0.8.1; RRID:SCR_001362; https://doi.org/10.5281/zenodo.8397156). Several confounding time-series were calculated based on the preprocessed BOLD: framewise displacement (FD) and three region-wise global signals. FD was computed using the Power formulation[89] (absolute sum of relative motions). The three global signals were extracted within the CSF, the WM, and the whole-brain masks. The confound time series derived from head motion estimates and global signals were expanded with the inclusion of temporal derivatives and quadratic terms for each.[90] Frames that exceeded a threshold of 0.2 mm FD were considered motion outliers. A Butterworth temporal filter extracted frequencies between 0.009 and 0.08 Hz. The band-pass filter was also applied to all confound regressors to avoid reintroducing filtered frequency bands.[91] The confound regression included the six head motion estimates and three region-wise global signals (CSF, WM, whole-brain), their temporal derivatives, quadratic terms, and the quadratic expansions of their derivatives (36 regressors).[90] We additionally included spike regressors associated with motion outliers. Following confound regression, the resulting residual resting-state timeseries were *z*-scored.

Population receptive field mapping. Stimuli were generated and presented using Python with the *PsychoPy* software package[92–94] and were viewed with MR-compatible goggles (NordicNeuroLab VisualSystem HD). The paradigm was adapted from the procedure described in Mackey et al.[32] Stimulus presentation was confined to a 16° $\times$ 16° field of view. The stimulus consisted of a bar aperture, which subtended 16° in length and subtended either 1°, 2°, or 3° in width. The use of different bar widths can aid the estimation of pRF size.[32] The width of the bar aperture was held constant within each run. Functional time-series of runs consisting of the same bar width were averaged prior to pRF modeling. The bar aperture comprised 3 equally sized rectangular patches of moving dots. Dot patches swept across the visual field in a discrete manner, changing location every 1.95 s (3 TRs). The step size of each change in location was 1°. There were eight possible sweep directions: left to middle, middle to left, right to middle, middle to right, bottom to middle, middle to bottom, top to middle, and middle to top. Each sweep consisted of 6 steps/trials (11.7 s) and was followed by a 0.65 s blank fixation interval (1 TR). Each patch spanned 5.1° along the side perpendicular to the sweep direction. A 0.35° gap separated each patch. Patches of width 1°, 2°, and 3° contained 100, 200, and 300 dots, respectively. Dots had a diameter of 0.093°, moved at 1.5 °/s, and updated their position 60 times per second. At each location, observers discriminated which of the two flanking patches contained dots moving in the same direction as the middle patch. Only one of the flanker patches moved in the same direction as the middle patch on each trial. Dot motion within the middle patch was always 100% coherent. Coherent dots moved along the length of the patch opposite to the sweep direction (left or right for vertical sweeps and up or down for horizontal sweeps). The coherence of the flanker patches' dot motion was staircased using a 1-up 3-down procedure.[95] Moving dots had a limited lifetime of 10 frames (167 ms). Each noise dot moved in a random direction for the extent of its lifetime. Participants completed 6-9 pRF runs (either 2 or 3 repeats of each patch width).

Spatial working memory paradigm. Spatial WM stimuli were generated and presented using Python with *PsychoPy* package.[92–94] Each trial consisted of a sample period (325 ms), a post-sample fixation period (325 ms), a delay period with an 'active' or 'drop' cue (8.45 s), and probe period (3.25 s). Participants were instructed to maintain their gaze on a small fixation dot (0.15°) throughout the sample and delay periods. During the sample period, participants were presented with a 0.75° diameter grayscale disc at 4° eccentricity. The polar angle of the stimulus was uniformly sampled from 6 bins (0°, 60°, 120°, 180°, 240°, or 300°). ±15° of angular jitter was added randomly to the polar angle on each trial to more finely sample polar-angle space. This jitter discouraged categorical encoding of the six nominal angle bins and improved estimation of continuous spatial tuning curves, while retaining a discretized structure that ensured balanced sampling of stimulus values. As a result, 186 out of a possible 360 polar angles could be presented. Participants were not told that stimuli were limited to specific ranges and were free to report any polar angle possible (0°–359°). 325 ms following the offset of the sample stimulus, the fixation dot turned either red or blue to indicate an 'active' or 'drop' trial, respectively. On drop trials, participants did not need to maintain the item in working memory and were instructed to make a random response when probed. Following the 8.45 s delay period, subjects were presented with a probe display (3.25 s). The probe display consisted of a disc stimulus that was identical to the sample stimulus (0.75° diameter disc at 4° eccentricity). Participants used their right index, middle, and ring fingers to respond. Holding the button with their index finger moved the disc counter-clockwise along possible stimulus locations at 4° eccentricity and the ring finger moved the disc clockwise. The middle finger flipped the probe 180°. The last recorded angle was taken as the memory report for that trial. The initial angle of the probe disc was randomly generated. The randomization of the probe disc was essential to ensure that subjects were unable to prospectively plan a specific motor action (i.e., clockwise versus counter-clockwise rotation) during the delay period. As a result, we can be confident that activity over the delay-period reflects visuospatial encoding of the stimulus as opposed to motor planning. The inter-trial interval (ITI) was jittered across trials and determined using optseq2 (https://surfer.nmr.mgh.harvard.edu/optseq). 10 run sequences were generated that maximized design matrix efficiency. We specified a minimum ITI of 4.55 s (7 TRs) and a maximum ITI of 12.35 s (19 TRs) and a total run duration of 263.9 s (406 TRs). Each fMRI run comprised 14 trials with 2 repeats of each angle bin (12 active trials + 2 drop trials). Subjects completed 10 runs during the session.

In Experiment 2, participants completed a modified version of the spatial WM task on a laptop immediately following TMS. The randomization of the probe stimulus was a necessary and intentional design decision to prevent prospective motor planning. However, this aspect of the design introduces response-stage noise (e.g., the sign of behavioral error is influenced by start position and direction of rotation). Therefore, the limited trial count in Experiment 1 leads to a behavioral readout that is conservative with respect to memory precision. The time course of Experiment 2 matched the structure of Experiment 1 but was optimized for higher trial counts within the window of the cTBS effect[39] by shortening the delay period to 1 s. Stimuli were presented at a viewing distance of approximately 60 cm. Each trial began with a 325 ms sample period in which two discs (0.75° diameter at 4° eccentricity) were shown simultaneously. One disc was red and the other was blue. After a 325 ms post-sample fixation interval, the central fixation dot changed color (blue or red) to cue which item should be retained across the delay. The probe display was identical in format to Experiment 1, and participants used the same three-button rotation response procedure to report the remembered polar angle. The initial orientation of the probe was again randomized on every trial to prevent prospective motor planning during the delay. The inter-trial interval was fixed at 0.5 s.

The task in Experiment 2 comprised 480 trials divided into 8 blocks of 60 trials, with brief self-paced breaks permitted between blocks. Trial identities were determined by a fully crossed design: six possible cued-item angle bins (0°, 60°, 120°, 180°, 240°, 300°), five angular offsets between cued and non-cued items (−120°, −60°, 60°, 120°, 180°), and two possible cued colors (red or blue). As in the fMRI experiment, ±15° of angular jitter was applied to both items on each trial, with identical jitter across the pair to preserve the specified offset.

pRF modeling. pRF analysis was performed using the analyzePRF MATLAB toolbox.[33] Voxel/vertex timeseries were modeled with a compressive spatial summation model,[33] which is an extension of the pRF model described by Dumoulin & Wandell.[96] This model includes an additional exponent parameter to account for subadditive spatial summation.[33] The model is formally expressed as:

$$r(t) = g \times \left[\iint S(x,y)G(x,y)dxdy\right]^n * h(t)$$

$$G(x,y) = e^{\frac{(x-x_0)^2 + (y-y_0)^2}{2\sigma^2}}$$

Where $r(t)$ is a voxel's predicted response at timepoint t, g is a gain parameter, S is a binary stimulus mask, G is a 2-dimensional isotropic Gaussian, n is an exponent parameter controlling the degree of subadditivity (≤1), $h(t)$ is a canonical hemodynamic response function, x_0 and y_0 are parameters defining the position of the Gaussian, and σ is a parameter defining the standard deviation of the Gaussian. Prior to fitting, functional time-series of runs consisting of the same experimental bar size (1, 2, or 3°) were averaged. To train the pRF model to predict responses to the spatial working memory task, we further averaged the time-series of each bar width across subjects.[35,97] The original pRF model described by Dumoulin and Wandell[96] involved a two-stage fitting procedure: an initial coarse grid-fit followed by an exhaustive non-linear optimization procedure using seed parameters from the grid-fit. However, recent studies have demonstrated that the full optimization does not outperform the grid-fit when predicting independent, left-out data.[32] It was argued that the grid-fit procedure is more robust to noise and better able to predict the responses of fronto-parietal voxels with large pRFs. We have previously used the grid-fit procedure to identify retinotopically organized cerebellar areas.[34] The grid-fitting procedure iterated over 49,045 possible parameter combinations (25 eccentricities, 24 angles, 17 widths, and 5 exponents) to find the grid point that produced the maximal correlation between the predicted and actual voxel timeseries. Predicted and actual timeseries were polynomial detrended and scaled to unit length prior to computing this correlation. The gain or amplitude parameter was estimated following the grid-fit procedure as:

$$a = corr(d, r) \times \frac{\|d\|}{\|r\|}$$

where d is the voxel timeseries, r is the predicted timeseries associated with the maximal correlation grid point, and $\| \|$ denotes vector length. Note that there is an interaction between the width of the Gaussian and the static nonlinearity parameter such that pRF size effectively equals $\frac{\sigma}{\sqrt{n}}$ as:

$$\left(e^{\frac{(x-x_0)^2+(y-y_0)^2}{2\sigma^2}}\right)^n = e^{\frac{(x-x_0)^2+(y-y_0)^2}{2\left(\frac{\sigma}{\sqrt{n}}\right)^2}}$$

TMS target definition. We first identified a cerebellar lobule VIIb target in each hemisphere. To do so, we projected the lobule VIIb retinotopic mask from the van Es cerebellar retinotopy atlas[35] into the subject's native T1 space. We then identified the voxel within this mask with the maximal pRF R^2 value, a pRF eccentricity <8°, and a polar angle preference ±45° from the ipsilateral hemifield horizontal meridian. If no voxel could be identified with these criteria, we removed the eccentricity criterion. If this also failed to identify a voxel within the lobule VIIb mask, we further removed the polar angle range criterion. Following the identification of a lobule VIIb target in each hemisphere, we created an 8 mm sphere centered on this same voxel and then applied a cerebellar cortical ribbon mask. This ROI was used as a seed in seed-based functional connectivity analyses to identify IPS, sPCS, and somatosensory control TMS targets. Due to the crossing of cortico-cerebellar circuits, left hemisphere cerebellar lobule VIIb was used as a seed for right hemisphere cerebral cortex targets, and right hemisphere cerebellar lobule VIIb was used as a seed for left hemisphere cerebral cortex targets. We defined anatomical search spaces for each cortical target. For the IPS target, we used the IPS0 label from the Wang probabilistic retinotopy atlas.[98] For the sPCS target, we combined the FEF and area 6a labels from the Glasser atlas.[84] Lastly, for the somatosensory control site, we combined the postcentral gyrus and paracentral gyrus/sulcus labels from the Destrieux atlas,[99] and then manually trimmed this label to lie medial to the hand knob portion of postcentral gyrus and posterior to the central sulcus. For each subject, these labels were projected from the *fsaverage* template surface to that subject's native surface (*mri_label2label*). The native surface label was then projected into native T1 volume space in a cortical ribbon constrained manner (*mri_label2vol*). IPS and sPCS targets were defined as the voxels with the maximal correlation with contralateral retinotopic cerebellar lobule VIIb. The somatosensory control target was identified as the voxel with the *minimum* absolute correlation with the contralateral cerebellar lobule VIIb retinotopic ROI.

TMS protocol. Repetitive TMS (rTMS) was delivered through a MagPro X100 magnetic stimulator and a 70 mm figure-8 coil (MC-B70, MagVenture). In an initial screen session, motor evoked potentials (MEPs) elicited using biphasic posterior-anterior stimulation and coil oriented 45° to the coronal plane were recorded from the left first dorsal interosseous (FDI) using surface electromyography (2-channel built-in EMG device, Rogue Research, Montreal). Active motor threshold (AMT) was recorded as the percentage of stimulator output that elicited an MEP of ≥ 200 μV peak-to-peak on five out of ten trials while the participant was contracting the FDI muscle at 20% of maximum.[100,101] At the end of the screening session, participants were given several pulses at 80% AMT over regions of the scalp roughly corresponding to cerebellar lobule VIIb, IPS, SPCS, and the control site to ensure that the stimulation intensity would be tolerable in future sessions. One participant reported discomfort related to stimulation of the approximate sPCS site and as a result only received lobule VIIb, IPS, and control stimulation in subsequent sessions. In the experimental sessions following the baseline fMRI session, we delivered continuous theta-burst stimulation (cTBS) with standard parameters: 3 pulses of stimulation at 50 Hz, repeated every 200 ms, for a total of 600 pulses over 40 seconds. We used the Brainsight Frameless neuronavigation system (Rogue Research, Montreal CA) to align the previously acquired structural scans to each participant's skull in real-time to allow for precise targeting.

TMS thresholds have been shown to depend on the distance between the coil and underlying target stimulation site.[102] To account for between-site differences in scalp-to-cortex distance, we adjusted stimulation intensity for each target relative to primary motor cortex (M1). Specifically, we (1) measured scalp-to-cortex distance at the M1 hand knob and at each target site (lobule VIIb, IPS, sPCS, and control) from each participant's T1-weighted anatomical image, and (2) computed a distance-adjusted active motor threshold as

$$AMT_{adj} = AMT + 3 \times (D_{site} - D_{M1})$$

where D_{site} and D_{M1} denote the scalp-to-cortex distance (cm) at the target site and the M1 hand knob, respectively. Continuous theta-burst stimulation was delivered at 80% of AMT_{adj} for each site.

Regions of Interest. ROIs were defined based on the baseline pRF mapping and atlas definitions. The anatomical search spaces for visual and parietal ROIs were extracted from a probabilistic retinotopy atlas (V1-V3, V3AB, IPS0-3).[98] We used the 'FEF' label from the HCP MMP atlas[84] for our sPCS ROI. A cerebellar retinotopic atlas was used to define the anatomical search space for lobule VIIb.[35] For visual cortex ROIs, we selected vertices with a pRF variance explained greater than 20% and an eccentricity between 3° and 8°. For the remaining ROIs in frontoparietal cortex and cerebellum, which possess larger receptive field sizes, we selected vertices with a pRF variance explained greater than 20% and no eccentricity criterion.

Spatial encoding model. To characterize the effect of TMS applied to each target on spatial selectivity throughout the brain, we employed a generative model-based, probabilistic decoding approach. This method was initially developed to characterize neural uncertainty associated with the encoding of orientation in visual cortex.[36] This method has been extended to model the encoding of motion direction[20] and spatial polar angle.[38] The model specifies a generative distribution, which represents the probability that a particular stimulus s will evoke an activation pattern b (i.e., $p(b|s)$). This distribution takes the form of a multivariate normal distribution with a stimulus-dependent mean vector $f(s)$ and a voxel covariance matrix Ω. The mean vector $f(s)$ is determined by each voxel's tuning profile over possible working memory stimulus locations (0°-359° polar angle at 4° eccentricity). Here, rather than use a linear channel-based approach as in prior work,[20,36–38,103] we instead use the nonlinear pRF model trained on the mapping data collected during the baseline fMRI session as our encoding model. The i-th voxel's spatial tuning curve was modeled as:

$$f_i(s) = a_i \times \left[\iint S(x,y,s)G_i(x,y)dxdy\right]^{n_i}$$

where a_i is the i-th voxel's gain estimated from the independent pRF mapping session, $S(x,y,s)$ is a binary stimulus mask of stimulus s (0.75° disc at 4° eccentricity, 0°–359° polar angle) spanning the entire 16° × 16° display, $G_i(x,y)$ is a 2D Gaussian representing i-th voxel's pRF, and n_i is the i-th voxel's estimated static nonlinearity parameter.

The regularized estimation of the noise covariance had two components: shrinkage of the sample correlation matrix towards a simpler structure comprising noise correlations related to tuning similarity and global noise shared across all voxels, as well as shrinkage of the sample variance toward its median value.

The noise correlation shrinkage target was constructed as follows. A sample covariance matrix was first computed from residuals resulting from the comparison of model-predicted responses ($f(s)$) and the training set, and converted to a correlation matrix:

$$\Omega_S = \frac{1}{t-1} X^T X$$

where X is a $t \times v$ matrix of residuals (where t is the number of trials in the training set and v is the number of voxel/vertices). The sample correlation matrix was then computed as:

$$R_S = (diag\ \Omega_S)^{-\frac{1}{2}}\, \Omega_S (diag\ \Omega_S)^{-\frac{1}{2}}$$

We then computed voxel tuning covariance as:

$$T = \frac{1}{p-1} W W^T$$

Where p is the number of display pixels, and W represents unwrapped voxel pRFs (2D Gaussians) for each voxel forming a $v \times p$ matrix. This covariance matrix was converted to a correlation matrix:

$$R_T = (diag\ T)^{-\frac{1}{2}}\, T\, (diag\ T)^{-\frac{1}{2}}$$

The global noise correlation was computed by regressing the empirical sample correlations onto tuning similarity and an intercept. The intercept captures the noise common to all voxels. The noise correlation shrinkage target was then computed as:

$$R_0 = aR_T + b\mathbf{1},\ (R_0)_{\boldsymbol{ii}} = 1$$

Where a is the slope parameter relating the tuning correlation matrix with the sample correlation matrix, b is the intercept, and $\mathbf{1}$ is a $v \times v$ matrix of ones. $(R_0)_{\boldsymbol{ii}} = 1$ ensures that the resulting matrix is a valid correlation matrix (ones along the diagonal). Sample correlations were then shrunk toward the target correlation matrix, with the degree of shrinkage controlled by λ.

$$\tilde{R} = \lambda R_0 + (1-\lambda) R_S$$

We additionally shrank voxel variance toward the median variance across voxels:

$$\widetilde{\omega}_i = \lambda_{var} median\ diag\ \Omega_S + (1-\lambda_{var})\omega_i$$

where λ_{var} is the parameter controlling the degree of shrinkage toward the median and ω_i is the sample variance of the i-th voxel. Thus, the noise covariance around voxel spatial tuning curves can be computed as:

$$\Omega = D^{\frac{1}{2}}\, \tilde{R}\, D^{\frac{1}{2}}$$

$$D = diag(\widetilde{\omega}_1, \dots, \widetilde{\omega}_v)$$

Thus, the generative distribution of voxel/vertex responses follows a multivariate distribution with a mean $f(s)$ and covariance Ω:

$$p(b|s;\theta) = N(f(s), \Omega)$$

where $\theta = \{x_0, y_0, \sigma, a, n, \lambda, \lambda_{var}\}$ are the model's parameters.

Model fitting and assessment was performed using a 10-fold leave-one-run-out cross-validation scheme, which iteratively partitioned the data into a training set and a test set. Trial-wise delay-period responses were estimated by averaging the final three TRs of the delay period, corresponding to the last 1.95 s before probe onset. This late-delay window was chosen to maximize separation from the sample-evoked response while avoiding contamination from probe- and response-related activity. Because the probe had not yet appeared during this window, these responses could not reflect processing of the probe stimulus or execution of the report. Prior to cross-validation, runs were reordered to align stimulus sequences across participants and then averaged trial-wise across participants to create a "super-subject." This approach follows prior large-scale pRF work[35,97] in which fMRI time series are aligned across subjects, averaged, and then modeled to recover stable group-level parameters. This procedure also has the advantage of allowing stimulation-order effects to be averaged across counterbalanced participants before estimating condition-specific spatial tuning. We assumed pRF centers (x_{0i}, y_{0i}) were fixed across tasks (spatial-attention motion-discrimination pRF task and spatial working-memory task) and TMS targets (CB, IPS, SPCS, control), but allowed voxel-wise modulation of baseline (additive), gain (multiplicative), and width (multiplicative) by task/target. Within each outer training fold, we performed a grid-based posterior approximation per voxel with log-uniform gain ∈ $[2^{-3}, 2^3]$ in 0.25-log2 steps, log-uniform width ∈ $[2^{-2}, 2^2]$ in 0.25-log2 steps, and baseline ∈ $[-0.20, 0.20]$ in 0.01 steps (17,425 combinations). Assuming flat priors over grid points, we converted log-densities to probabilities, normalized, and took the posterior mean of each parameter to regenerate $f(s)$ for $0° - 359°$.

We then performed an inner 9-fold cross-validation to estimate noise covariance hyperparameters (λ, λ_{var}). This procedure comprised a coarse grid search ($\lambda \in (0,1]$ and $\lambda_{var} \in [0, 1]$) followed by a fine grid search around the point associated with the minimum loss.[104] At each grid point, we evaluated the negative log-likelihood (scaled by 2/p) of the validation sample covariance (S_{val}) under a zero-mean Gaussian model with covariance Ω:[105]

$$\mathcal{L}(\Omega, S_{val}) = \frac{1}{p}[tr(\Omega^{-1}S_{val}) + \ln\det\Omega]$$

where constant terms independent of Ω have been dropped.[105] Hyperparameters associated with the minimum cross-validated loss were then used to re-estimate Ω for the entire training set (9 runs).

Then, for each trial in the test set, we used Bayes rule to obtain a posterior probability distribution over spatial polar angle (0°–359°), indicating which spatial location was most probable given the observed pattern of BOLD responses.

$$p(s|b;\hat{\theta}) \propto p(b|s;\hat{\theta})p(s)$$

Where $\hat{\theta}$ are the estimated model parameters. The stimulus prior, $p(s)$, was set to 1 for all polar angles as experimental stimuli were uniformly distributed. The posterior mean represented the decoded location and the posterior standard deviation represented the decoded uncertainty. Using the posterior mean, we calculated mean circular squared error (MSE_{circ}). The inverse of MSE_{circ} produced a measure of estimation efficiency, with higher values associated with more accurate estimation of spatial location.[106] To test for differences in decoded uncertainty, we computed the mean posterior standard deviation across trials for each ROI and stimulation condition.

Tuning curve modulation. To identify which aspects of voxel spatial tuning best account for the observed decoding effects, we performed a post-hoc "tuning-curve modulation" analysis that estimates per-voxel modulation of gain (multiplicative), width (multiplicative scaling of pRF size), baseline response (additive), and response variance (multiplicative) across TMS conditions. This analysis uses the same encoding model and notation introduced above for spatial decoding, but is conducted outside the cross-validation procedure. The center coordinates (x_{0i}, y_{0i}) of each voxel's pRF and the power-law nonlinearity parameter (n_i) were held fixed across tasks and TMS sessions.

We modeled the session-dependent modulation of voxel tuning curves with a parameter vector

$$\phi_i = \left(\phi_{size},\ \phi_{gain},\ \phi_{baseline},\ \phi_{var}\right),$$

where ϕ_{size} multiplicatively scales the pRF width, ϕ_{gain} multiplicatively scales gain, $\phi_{baseline}$ additively scales the baseline, and ϕ_{var} multiplicatively scales the voxel's response variance. For each voxel i, we generated a library of candidate tuning curves by scaling the pRF width in 2D display space and then projecting to polar angle. Specifically, we replaced σ_i with $\phi_{size}\,\sigma_i$ inside the Gaussian $G_i(x, y)$ defined over the 16° × 16° display, evaluated the 2D integral with the stimulus mask $S(x, y, s)$ to obtain $f_i^{(\phi_{size})}(s)$. In contrast, the remaining modulation parameters operated on the resulting 1D tuning curve across polar angle. The modulated tuning curve used in the likelihood was

$$\tilde{f}_i(s \mid \phi_i) = \phi_{gain} f_i^{(\phi_{size})}(s) + \phi_{baseline},$$

with stimulus-independent variance $\phi_{var}\hat{v}_i$ (where $\hat{v}_i$ is the i^{th} voxel's variance estimate from the baseline pRF-mapping residuals).

Let r_{ti} denote the measured delay-period response of voxel i on working memory trial t, and s_t the presented polar angle stimulus on that trial. Assuming conditionally independent Gaussian noise,

$$p(r_{ti} \mid \phi_i) = \prod_t \mathcal{N}\left(r_{ti} \,\middle|\, \tilde{f}_i(s_t \mid \phi_i),\ \phi_{var}\hat{v}_i\right),$$

We evaluated the log-likelihood on a dense Cartesian grid with uniform priors:

$$\phi_{size} \in 2^{[-2,\,2]}\ (0.25\ log_2\ steps),$$
$$\phi_{gain} \in 2^{[-3,\,3]}\ (0.25\ log_2\ steps),$$
$$\phi_{var} \in 2^{[-3,\,3]}\ (0.25\ log_2\ steps),$$
$$\phi_{baseline} \in [-0.2, 0.2]\ (0.01\ steps)$$

Posterior weights were obtained by exponentiating log-likelihoods after subtracting the maximum log-likelihood to avoid numerical underflow, and then normalizing to sum to one. With uniform priors on the grid, posterior means for each parameter were computed as weighted averages under these normalized weights. We then regenerated each voxel's modulated tuning curve across polar-angle,

$$\tilde{f}_i(s) = \bar{\phi}_{gain} f_i^{(\bar{\phi}_{size})}(s) + \bar{\phi}_{baseline},$$

with $\bar{\phi}_{var}\, \hat{v}_i$ as the response variance.

Information theoretic analysis. To complement the decoding-based analysis of expressed spatial precision, we next quantified how much spatial information is represented in the population code itself, independent of trial-by-trial decoding variability. Our decoding analysis assesses how accurately a remembered location can be reconstructed from trialwise observed activation patterns, as well as provides an estimate of the uncertainty of that reconstruction. On the other hand, Fisher information (FI) characterizes the intrinsic representational capacity of the population relative to its noise covariance.[107–111] FI provides a measure of the precision theoretically available to an ideal observer, defining the upper bound on how accurately small changes in stimulus location can be discriminated from the population response.[107,112,113]

For a population response $f(s)$ with noise covariance Σ, FI is

$$I(s) = f'(s)^T\, \Sigma^{-1}\, f'(s),$$

where $f'(s)$ is the derivative of the population tuning curve with respect to the stimulus. Higher FI indicates greater precision for discriminating small changes in stimulus value (here, polar angle).

We adopted a cross-validated implementation of FI to minimize bias from overfitting and shared noise structure.[114] In conventional formulations, the same data are used to estimate both tuning functions and noise covariance, inflating information estimates. Here, the tuning functions were estimated from the training runs, while the noise covariance was estimated from the held-out test run, ensuring that the signal and noise terms were statistically independent and yielding an unbiased estimate of representational precision.

In each fold k of the ten-fold cross-validation, we formed the population response vector $\tilde{f}^{(k)}(s)$ by stacking the modulated voxel tuning functions $\tilde{f}_i^{(k)}(s)$ across voxels over polar angle $s \in \{0°, \ldots, 359°\}$. The functions $\tilde{f}_i^{(k)}(s)$ were estimated from nine training runs and then evaluated on the held-out run to compute prediction residuals for covariance estimation. These curves reflected modulation of gain, width, and baseline but omitted the variance-modulation term (ϕ_{var}), so that noise was captured directly from independent test-run residuals. The FI analysis therefore isolates changes in information content arising from the shape of the population tuning functions.

Because the number of test trials per fold (T) is small relative to the number of voxels (n), the sample covariance in voxel space is singular or poorly conditioned. To obtain a stable, invertible covariance estimate that emphasizes signal-dominant directions, responses were mean-centered across polar angles and projected into a low-dimensional signal subspace obtained from the right singular vectors $V^{(k)} \in \mathbb{R}^{n \times r}$ of the centered tuning matrix, retaining the minimal r

that explained at least 99% of across-angle variance (thereby ensuring $r \ll T$). All subsequent computations were performed in this subspace:

$$\tilde{f}_{pca}^{(k)}(s) = \tilde{f}_0^{(k)}(s)\, V^{(k)}.$$

Derivatives with respect to polar angle were computed by circular first differences with step $\Delta\theta = \pi/180$ radians, and then projected into the same subspace:

$$\tilde{f}_{pca}^{(k)\,\prime}(s) = \tilde{f}^{(k)\,\prime}(s)\, V^{(k)}.$$

On the left-out test run for fold k, trial-wise responses were mean-centered using the same voxel means and projected into the same subspace. Predicted responses at the presented angles were subtracted to form residuals in the subspace, and a covariance matrix $C^{(k)}$ was computed from these residuals. Fisher information at angle s for fold k was then

$$I_k(s) = \tilde{f}_{pca}^{(k)\,\prime}(s)^T\, C^{(k)\,-1}\, \tilde{f}_{pca}^{(k)\,\prime}(s) \times (\frac{T-r-2}{T}),$$

where $(T-r-2)/T$ applies the inverse-Wishart finite-sample correction for estimating $C^{(k)\,-1}$ in an r-dimensional subspace.[115] The cross-validated FI curve was obtained by averaging across folds,

$$I_{cv}(s) = \frac{1}{K}\sum_{k=1}^{K} I_k(s)\,,$$

and a scalar summary for each ROI was the mean across polar angles,

$$\bar{I}_{cv} = \frac{1}{360}\sum_{s=0}^{359} I_{cv}(s)$$

Uncertainty in $\bar{I}_{cv}$ was estimated with a bootstrap over runs. In each of 5,000 iterations, folds were resampled with replacement, $\bar{I}_{cv}$ recomputed, and the mean FI re-estimated.

Statistical Analysis. All analyses were implemented in R (Version 4.5.1). All comparisons used non-parametric permutation tests with two-sided p-values from 5,000 permutations. Unless noted, contrasts were made relative to the somatosensory control site. False-discovery rate (Benjamini–Hochberg) corrections were applied once per family: 3 comparisons for behavioral analyses (CB, IPS, or sPCS versus Control) and 30 comparisons for fMRI analyses (10 ROIs × 3 session contrasts). Permutation p-values were computed as $(\#\{|\,T_{\text{perm}}\,| \geq |\,T_{\text{obs}}\,|\} + 1)/(N_{\text{perm}} + 1)$[116] and then FDR-adjusted.

To test for differences in spatial working-memory recall error across TMS conditions, trials were first screened to remove likely guesses as on random-guess trials the error distribution is approximately uniform, so error magnitude ceases to index precision (e.g., $45°$ versus $90°$ error both reflect failed encoding for a guess rather than graded loss of precision). In Experiment 1, trials with absolute error greater than the mean absolute error plus two standard deviations (computed within subject and session) were excluded (Control: 4.17% ± 0.73% of trials excluded; CB: 4.44% ± 0.85%; IPS: 2.62 ± 0.32%; sPCS: 5.28% ± 0.88%). In Experiment 2, the same error criterion was applied and trials with reaction time > mean RT plus two standard deviations (reaction time was not recorded in Experiment 1) were additionally excluded (Control: 7.15% ± 0.57% of trials excluded; CB: 6.74% ± 0.56%; IPS: 6.36% ± 0.61%; sPCS: 6.81% ± 0.36%). Session effects were then tested with a hierarchical paired (within-subject) permutation procedure in which, for each contrast, session labels were randomly swapped within subject, subject-level means recomputed, and the across-subject mean paired difference used as the test statistic. To account for potential sequence effects, we additionally regressed out session order using a robust linear mixed-effects model with a random intercept per subject (*robustlmm* package),[117] and repeated the permutation test on the residualized errors.

Differences in decoding metrics were assessed by permuting the predicted polar angle across trials before computing estimation efficiency (inverse mean circular squared error) within each ROI and session. Decoded uncertainty (mean posterior standard deviation) was tested by permuting session labels at the trial level within ROI before recomputing session means and taking the difference between each experimental session (CB, IPS, and sPCS) and the control session.

Session effects in tuning-curve modulation were tested by permuting session labels across voxels within an ROI for each posterior-mean modulation parameter (gain, baseline, pRF width, and variance) and recomputing the difference between each stimulation condition and control stimulation.

To test for differences in Fisher information across sessions, we used the ten cross-validated run-level values per session (each averaged over polar angle). Session labels were permuted across runs within ROI to generate null distributions for the mean difference between each stimulation condition and control.

Data Availability
Summary behavioral and fMRI datasets underlying the statistical analyses reported in this study will be publicly available through the Open Science Framework (OSF) at https://osf.io/2b6um. Owing to the large size of the raw MRI datasets and associated storage constraints, raw neuroimaging data are not included in the public repository but may be made available by the corresponding author upon reasonable request.

Code Availability
Code used for modeling, statistical analysis, and figure generation will be made publicly available at https://github.com/brissend/CB_WM_TMS.

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Author Contributions Statement
J.A.B. and T.G.L. conceived the study. J.A.B. collected the data. J.A.B. carried out data analysis and wrote the original draft of the manuscript. All authors reviewed the manuscript and provided critical revisions. T.G.L. and M.V. provided resources and supervision.

Competing Interests Statement
The authors declare no competing interests.

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Figure Legends/Captions

Figure 1. Spatial working memory (WM) paradigms and stimulation induced recall deficits. (a) Inside an fMRI scanner (Experiment 1), participants were presented with a disc stimulus (0.75°) at 4° eccentricity and were instructed to remember the location. 325 ms following the offset of the stimulus presentation period, participants were cued as to whether the trial was an 'active' or 'drop' trial (120 active trials / 20 drop trials). On 'active' trials, participants were instructed to maintain the location of the sample stimulus over the subsequent 8.45 s delay period, while on 'drop' trials they were instructed to drop the stimulus from memory and make a random response. The long delay period allowed us to isolate the BOLD response associated with WM maintenance. Following the delay period, participants were given 3.25 s to adjust a probe disc stimulus to match the remembered location. The initial angle of the probe stimulus was randomized with respect to the memorized location, which ensured that participants were unable to prospectively plan their upcoming motor response (i.e., direction of rotation) during the delay period. Thus, delay-period activity can be inferred to relate to visuospatial WM maintenance rather than preparatory motor activity. (b) In a follow-up psychophysics experiment, participants performed a similar WM task with a larger set size (2 items) and a shorter delay-period (1 s; 480 total trials). A retroactive cue presented 325 ms following the offset of the sample display indicated which of the two stimuli was to be maintained over the delay period. The probe stimulus was again randomized to prevent prospective motor planning. (c,d) Cortico-cerebellar TMS produces WM recall deficits. Parietal stimulation reliably increased recall error in the fMRI experiment (c), while cerebellar, parietal, and frontal stimulation increased recall error in the higher-powered psychophysics experiment (d).

Figure 2. Population receptive fielding (pRF) mapping. (a) In an initial fMRI session, participants performed an attentionally demanding visual field mapping task which required discriminating and comparing the direction of motion of three equal sized patches of moving dot patterns at different positions within the display.[32] Resulting BOLD responses were modeled with compressive spatial summation model.[33] This model produces a 2D Gaussian receptive field for each voxel (x, y, σ) along with gain (a) and compressive nonlinearity (n) parameters. (b) Polar angle preferences across cortex and cerebellum. These results replicate prior work demonstrating spatial selectivity in cerebellum.[34,35] (c) Eccentricity preferences across cortex and cerebellum.

Figure 3. Probabilistic spatial encoding model. (a) fMRI data were modeled with a spatial encoding framework that predicted voxelwise BOLD responses from population receptive field parameters derived from the baseline mapping session (Fig. 2a) and stimulus masks corresponding with all possible spatial WM locations probed in the TMS-fMRI sessions. (b) Spatial tuning curve modulation for an example voxel. The model accounted for session-specific modulation of spatial tuning curves. Each black dot represents the late-delay BOLD response for a single WM trial. The blue curve represents the predicted response to each spatial location from the original model parameters, while the red curve reflects an adjustment in gain, width,

and baseline activation as a function of task. (c) Probabilistic decoding of spatial location. For each trial, we computed the posterior probability of each possible stimulus location given the observed multi-voxel BOLD activity on that trial, while taking into account covariance across voxels (see methods).

Figure 4. Cortico-cerebellar stimulation degrades spatial decoding across multiple brain regions. (a) Mean posterior probability distribution. Averaging trial-aligned posterior distributions produces a session-level probability density over stimulus locations for each region of interest (ROI). Sharper peaks around 0° reflect better decoding of the remembered spatial location. (b) Estimation efficiency (inverse of mean circular squared error between the posterior mean and the true stimulus) reflecting decoding accuracy for each session and region. (c) Mean posterior standard deviation reflecting decoding uncertainty. Error bars reflect a bootstrap estimate of standard error. Horizontal bars indicate corrected $p < 0.05$ from a hierarchical permutation test. Each column reflects results for one ROI. IPS – intraparietal sulcus; sPCS – superior pre-central sulcus.

Figure 5. Spatial tuning curve modulation and hypothetical effects of stimulation. (a) Analysis separately quantified TMS-induced changes in multiplicative gain (ϕ_{gain}), tuning width (ϕ_{size}), additive baseline shifts ($\phi_{baseline}$), and residual variance (ϕ_{var}) relative to the baseline pRF-predicted response. Gaussian center (x, y) and compressive nonlinearity exponent (n) were held constant. (b) Possible effects of stimulation on different aspects of voxel spatial tuning.

Figure 6. TMS induced changes in spatial tuning during WM maintenance. (a) Mean tuning curve across voxels for each stimulation session within each ROI. For visualization, the tuning curves are demeaned across sessions. Shaded ribbons reflect bootstrap estimate of SEM across voxels. (b) Spatial tuning curve baseline for each session and ROI. This value reflects stimulus-independent BOLD activity during the delay-period. (c) Spatial tuning curve gain for each session and ROI. This value reflects the BOLD response for preferred stimulus locations relative to non-preferred locations. (d) Spatial tuning curve width for each session and ROI. (e) Spatial tuning curve variance for each session and ROI reflecting the distribution of BOLD responses for repeated presentations of the same stimulus. (f) Cross-validated Fisher information for each session and ROI. This value reflects how sensitively a region's mean response varies with spatial location relative to its noise covariance, thus characterizing the intrinsic representational capacity supported by the underlying population code. This measure represents the net impact of all TMS-induced changes in spatial tuning. Error bars across b-f reflect a bootstrap estimate of SEM. Horizontal bars indicate corrected $p < 0.05$ from a hierarchical permutation test.

1. D'Esposito, M. & Postle, B. R. The Cognitive Neuroscience of Working Memory. *Annu Rev Psychol* **66**, 1–28 (2015).

2. Fuster, J. M. & Alexander, G. E. Neuron Activity Related to Short-Term Memory. *Science* **173**, 652–654 (1971).

3. Curtis, C. E. & D'Esposito, M. Persistent activity in the prefrontal cortex during working memory. *Trends Cogn. Sci.* **7**, 415–423 (2003).

4. Serences, J. T. Neural mechanisms of information storage in visual short-term memory. *Vision Res* **128**, 53–67 (2016).

5. Sprague, T. C., Ester, E. F. & Serences, J. T. Reconstructions of Information in Visual Spatial Working Memory Degrade with Memory Load. *Curr Biol* **24**, 2174–2180 (2014).

6. Ester, E. F., Sprague, T. C. & Serences, J. T. Parietal and Frontal Cortex Encode Stimulus-Specific Mnemonic Representations during Visual Working Memory. *Neuron* **87**, 893–905 (2015).

7. Serences, J. T., Ester, E. F., Vogel, E. K. & Awh, E. Stimulus-Specific Delay Activity in Human Primary Visual Cortex. *Psychol Sci* **20**, 207–214 (2008).

8. Curtis, C. E. & Sprague, T. C. Persistent Activity During Working Memory From Front to Back. *Front. Neural Circuits* **15**, 696060 (2021).

9. Christophel, T. B., Klink, P. C., Spitzer, B., Roelfsema, P. R. & Haynes, J.-D. The Distributed Nature of Working Memory. *Trends Cogn Sci* **21**, 111–124 (2017).

10. Michalka, S. W., Kong, L., Rosen, M. L., Shinn-Cunningham, B. G. & Somers, D. C. Short-Term Memory for Space and Time Flexibly Recruit Complementary Sensory-Biased Frontal Lobe Attention Networks. *Neuron* **87**, 882–892 (2015).

11. Lee, T. G. & D'Esposito, M. The Dynamic Nature of Top-Down Signals Originating from Prefrontal Cortex: A Combined fMRI–TMS Study. *J Neurosci* **32**, 15458–15466 (2012).

12. Postle, B. R. Working memory as an emergent property of the mind and brain. *Neuroscience* **139**, 23–38 (2006).

13. Funahashi, S., Bruce, C. J. & Goldman-Rakic, P. S. Mnemonic coding of visual space in the monkey's dorsolateral prefrontal cortex. *J Neurophysiol* **61**, 331–349 (1989).

14. Goldman-Rakic, P. S. Cellular basis of working memory. *Neuron* **14**, 477–485 (1995).

15. Chafee, M. V. & Goldman-Rakic, P. S. Matching Patterns of Activity in Primate Prefrontal Area 8a and Parietal Area 7ip Neurons During a Spatial Working MemoryTask. *J Neurophysiol* **79**, 2919–2940 (1998).

16. Harrison, S. A. & Tong, F. Decoding reveals the contents of visual working memory in early visual areas. *Nature* **458**, 632–635 (2009).

17. Ester, E. F., Serences, J. T. & Awh, E. Spatially Global Representations in Human Primary Visual Cortex during Working Memory Maintenance. *J. Neurosci.* **29**, 15258–15265 (2009).

18. Riggall, A. C. & Postle, B. R. The Relationship between Working Memory Storage and Elevated Activity as Measured with Functional Magnetic Resonance Imaging. *J Neurosci* **32**, 12990–12998 (2012).

19. Christophel, T. B., Hebart, M. N. & Haynes, J.-D. Decoding the Contents of Visual Short-Term Memory from Human Visual and Parietal Cortex. *J Neurosci* **32**, 12983–12989 (2012).

20. Brissenden, J. A., Tobyne, S. M., Halko, M. A. & Somers, D. C. Stimulus-Specific Visual Working Memory Representations in Human Cerebellar Lobule VIIb/VIIIa. *J Neurosci* JN-RM-1253-20 (2021) doi:10.1523/jneurosci.1253-20.2020.

21. Diedrichsen, J., Verstynen, T., Schlerf, J. & Wiestler, T. Advances in functional imaging of the human cerebellum. *Curr Opin Neurol* **23**, 382–387 (2010).

22. Shahshahani, L., King, M., Nettekoven, C., Ivry, R. B. & Diedrichsen, J. Selective recruitment of the cerebellum evidenced by task-dependent gating of inputs. *eLife* **13**, RP96386 (2024).

23. Miall, R. C. & Wolpert, D. M. Forward Models for Physiological Motor Control. *Neural Networks* **9**, 1265–1279 (1996).

24. Wolpert, D. M., Miall, R. C. & Kawato, M. Internal models in the cerebellum. *Trends Cogn Sci* **2**, 338–347 (1998).

25. Ito, M. Control of mental activities by internal models in the cerebellum. *Nat Rev Neurosci* **9**, 304–313 (2008).

26. Raymond, J. L. & Medina, J. F. Computational Principles of Supervised Learning in the Cerebellum. *Annu. Rev. Neurosci.* **41**, 233–253 (2018).

27. Shadmehr, R. Population coding in the cerebellum: a machine learning perspective. *J Neurophysiol* **124**, 2022–2051 (2020).

28. Proville, R. D. *et al.* Cerebellum involvement in cortical sensorimotor circuits for the control of voluntary movements. *Nat. Neurosci.* **17**, 1233–1239 (2014).

29. Gao, Z. *et al.* A cortico-cerebellar loop for motor planning. *Nature* **563**, 113–116 (2018).

30. Chabrol, F. P., Blot, A. & Mrsic-Flogel, T. D. Cerebellar Contribution to Preparatory Activity in Motor Neocortex. *Neuron* **103**, 506-519.e4 (2019).

31. Pemberton, J., Chadderton, P. & Costa, R. P. Cerebellar-driven cortical dynamics can enable task acquisition, switching and consolidation. *Nat. Commun.* **15**, 10913 (2024).

32. Mackey, W. E., Winawer, J. & Curtis, C. E. Visual field map clusters in human frontoparietal cortex. *Elife* **6**, e22974 (2017).

33. Kay, K. N., Winawer, J., Mezer, A. & Wandell, B. A. Compressive spatial summation in human visual cortex. *J Neurophysiol* **110**, 481–494 (2013).

34. Brissenden, J. A. *et al.* Topographic Cortico-cerebellar Networks Revealed by Visual Attention and Working Memory. *Curr Biol* **28**, 3364-3372.e5 (2018).

35. Es, D. M. van, Zwaag, W. van der & Knapen, T. Topographic Maps of Visual Space in the Human Cerebellum. *Curr Biol* **29**, 1689-1694.e3 (2019).

36. Bergen, R. S. van, Ma, W. J., Pratte, M. S. & Jehee, J. F. M. Sensory uncertainty decoded from visual cortex predicts behavior. *Nat Neurosci* **18**, 1728–1730 (2015).

37. Geurts, L. S., Cooke, J. R. H., Bergen, R. S. van & Jehee, J. F. M. Subjective confidence reflects representation of Bayesian probability in cortex. *Nat. Hum. Behav.* **6**, 294–305 (2022).

38. Li, H.-H., Sprague, T. C., Yoo, A. H., Ma, W. J. & Curtis, C. E. Joint representation of working memory and uncertainty in human cortex. *Neuron* (2021) doi:10.1016/j.neuron.2021.08.022.

39. Huang, Y.-Z., Edwards, M. J., Rounis, E., Bhatia, K. P. & Rothwell, J. C. Theta Burst Stimulation of the Human Motor Cortex. *Neuron* **45**, 201–206 (2005).

40. Franca, M., Koch, G., Mochizuki, H., Huang, Y.-Z. & Rothwell, J. C. Effects of theta burst stimulation protocols on phosphene threshold. *Clin. Neurophysiol.* **117**, 1808–1813 (2006).

41. Silvanto, J., Bona, S. & Cattaneo, Z. Initial activation state, stimulation intensity and timing of stimulation interact in producing behavioral effects of TMS. *Neuroscience* **363**, 134–141 (2017).

42. Koch, G. *et al.* Changes in intracortical circuits of the human motor cortex following theta burst stimulation of the lateral cerebellum. *Clin. Neurophysiol.* **119**, 2559–2569 (2008).

43. Popa, T., Russo, M. & Meunier, S. Long-lasting inhibition of cerebellar output. *Brain Stimul.* **3**, 161–169 (2010).

44. Popa, D. *et al.* Functional Role of the Cerebellum in Gamma-Band Synchronization of the Sensory and Motor Cortices. *J. Neurosci.* **33**, 6552–6556 (2013).

45. Popa, T. *et al.* Cerebellar Processing of Sensory Inputs Primes Motor Cortex Plasticity. *Cereb. Cortex* **23**, 305–314 (2013).

46. Bologna, M. *et al.* Effects of cerebellar continuous theta burst stimulation on resting tremor in Parkinson's disease. *Park. Relat. Disord.* **21**, 1061–1066 (2015).

47. Mackey, W. E., Devinsky, O., Doyle, W. K., Meager, M. R. & Curtis, C. E. Human Dorsolateral Prefrontal Cortex Is Not Necessary for Spatial Working Memory. *J Neurosci* **36**, 2847–2856 (2016).

48. Mackey, W. E., Devinsky, O., Doyle, W. K., Golfinos, J. G. & Curtis, C. E. Human parietal cortex lesions impact the precision of spatial working memory. *J Neurophysiol* **116**, 1049–54 (2016).

49. Chen, S. H. A. & Desmond, J. E. Cerebrocerebellar networks during articulatory rehearsal and verbal working memory tasks. *Neuroimage* **24**, 332–338 (2005).

50. Chen, S. H. A. & Desmond, J. E. Temporal dynamics of cerebro-cerebellar network recruitment during a cognitive task. *Neuropsychologia* **43**, 1227–1237 (2005).

51. Kirschen, M. P., Chen, S. H. A., Schraedley-Desmond, P. & Desmond, J. E. Load- and practice-dependent increases in cerebro-cerebellar activation in verbal working memory: an fMRI study. *Neuroimage* **24**, 462–472 (2005).

52. Hayter, A. L., Langdon, D. W. & Ramnani, N. Cerebellar contributions to working memory. *Neuroimage* **36**, 943–954 (2007).

53. Allen, G., Buxton, R. B., Wong, E. C. & Courchesne, E. Attentional Activation of the Cerebellum Independent of Motor Involvement. *Science* **275**, 1940–1943 (1997).

54. Thürling, M. *et al.* Involvement of the cerebellar cortex and nuclei in verbal and visuospatial working memory: A 7T fMRI study. *NeuroImage* **62**, 1537–1550 (2012).

55. Sobczak-Edmans, M. *et al.* Temporal dynamics of visual working memory. *NeuroImage* **124**, 1021–1030 (2016).

56. Brissenden, J. A., Levin, E. J., Osher, D. E., Halko, M. A. & Somers, D. C. Functional Evidence for a Cerebellar Node of the Dorsal Attention Network. *J Neurosci Official J Soc Neurosci* **36**, 6083–96 (2016).

57. Brissenden, J. A. & Somers, D. C. Cortico-Cerebellar Networks for Visual Attention and Working Memory. *Curr Opin Psychology* **29**, 239–247 (2019).

58. Stoodley, C. J. & Schmahmann, J. D. Functional topography in the human cerebellum: A meta-analysis of neuroimaging studies. *Neuroimage* **44**, 489–501 (2009).

59. Stoodley, C. J., Valera, E. M. & Schmahmann, J. D. Functional topography of the cerebellum for motor and cognitive tasks: An fMRI study. *Neuroimage* **59**, 1560–1570 (2012).

60. Buckner, R. L., Krienen, F. M., Castellanos, A., Diaz, J. C. & Yeo, B. T. T. The organization of the human cerebellum estimated by intrinsic functional connectivity. *J Neurophysiol* **106**, 2322–2345 (2011).

61. Guell, X., Schmahmann, J. D., Gabrieli, J. D. & Ghosh, S. S. Functional gradients of the cerebellum. *Elife* **7**, e36652 (2018).

62. King, M., Hernandez-Castillo, C. R., Poldrack, R. A., Ivry, R. B. & Diedrichsen, J. Functional boundaries in the human cerebellum revealed by a multi-domain task battery. *Nat Neurosci* **22**, 1371–1378 (2019).

63. Nettekoven, C. *et al.* A hierarchical atlas of the human cerebellum for functional precision mapping. *Nat. Commun.* **15**, 8376 (2024).

64. Romano, V. *et al.* Stage-dependent cerebrocerebellar communication during sensorimotor processing. *Nat. Commun.* **16**, 8812 (2025).

65. Imamizu, H. *et al.* Human cerebellar activity reflecting an acquired internal model of a new tool. *Nature* **403**, 192–195 (2000).

66. Morton, S. M. & Bastian, A. J. Cerebellar Contributions to Locomotor Adaptations during Splitbelt Treadmill Walking. *J Neurosci* **26**, 9107–9116 (2006).

67. Bastian, A. J. Learning to predict the future: the cerebellum adapts feedforward movement control. *Curr. Opin. Neurobiol.* **16**, 645–649 (2006).

68. Tseng, Y., Diedrichsen, J., Krakauer, J. W., Shadmehr, R. & Bastian, A. J. Sensory Prediction Errors Drive Cerebellum-Dependent Adaptation of Reaching. *J. Neurophysiol.* **98**, 54–62 (2007).

69. Schlerf, J., Ivry, R. B. & Diedrichsen, J. Encoding of Sensory Prediction Errors in the Human Cerebellum. *J. Neurosci.* **32**, 4913–4922 (2012).

70. Tanaka, H., Ishikawa, T., Lee, J. & Kakei, S. The Cerebro-Cerebellum as a Locus of Forward Model: A Review. *Front. Syst. Neurosci.* **14**, 19 (2020).

71. Brissenden, J. A., Yin, Y., Vesia, M. & Lee, T. G. Errors of attention adaptively warp spatial cognition. *Nat. Hum. Behav.* 1–12 (2025) doi:10.1038/s41562-025-02109-5.

72. Sreenivasan, K. K., Curtis, C. E. & D'Esposito, M. Revisiting the role of persistent neural activity during working memory. *Trends Cogn. Sci.* **18**, 82–89 (2014).

73. Xu, Y. Sensory Cortex Is Nonessential in Working Memory Storage. *Trends Cogn. Sci.* **22**, 192–193 (2018).

74. Adam, K. C. S., Rademaker, R. L. & Serences, J. T. Visual Memory. 5–25 (2022) doi:10.4324/9781003158134-2.

75. Esteban, O. *et al.* fMRIPrep: a robust preprocessing pipeline for functional MRI. *Nat Methods* **16**, 111–116 (2019).

76. Gorgolewski, K. *et al.* Nipype: A Flexible, Lightweight and Extensible Neuroimaging Data Processing Framework in Python. *Front. Neuroinformatics* **5**, 13 (2011).

77. Dickie, E. W. *et al.* Ciftify: A framework for surface-based analysis of legacy MR acquisitions. *NeuroImage* **197**, 818–826 (2019).

78. Tustison, N. J. *et al.* N4ITK: Improved N3 Bias Correction. *IEEE Trans. Méd. Imaging* **29**, 1310–1320 (2010).

79. Avants, B. B., Epstein, C. L., Grossman, M. & Gee, J. C. Symmetric diffeomorphic image registration with cross-correlation: Evaluating automated labeling of elderly and neurodegenerative brain. *Méd. Image Anal.* **12**, 26–41 (2008).

80. Zhang, Y., Brady, M. & Smith, S. Segmentation of Brain MR Images Through a Hidden Markov Random Field Model and the Expectation-Maximization Algorithm. *IEEE Trans. Méd. Imaging* **20**, 45 (2001).

81. Dale, A. M., Fischl, B. & Sereno, M. I. Cortical Surface-Based Analysis I. Segmentation and Surface Reconstruction. *NeuroImage* **9**, 179–194 (1999).

82. Klein, A. *et al.* Mindboggling morphometry of human brains. *PLoS Comput. Biol.* **13**, e1005350 (2017).

83. Robinson, E. C. *et al.* Multimodal surface matching with higher-order smoothness constraints. *NeuroImage* **167**, 453–465 (2018).

84. Glasser, M. F. *et al.* A multi-modal parcellation of human cerebral cortex. *Nature* **536**, 171–178 (2016).

85. Cox, R. W. & Hyde, J. S. Software tools for analysis and visualization of fMRI data. *NMR Biomed.* **10**, 171–178 (1997).

86. Greve, D. N. & Fischl, B. Accurate and robust brain image alignment using boundary-based registration. *NeuroImage* **48**, 63–72 (2009).

87. Jenkinson, M., Bannister, P., Brady, M. & Smith, S. Improved Optimization for the Robust and Accurate Linear Registration and Motion Correction of Brain Images. *NeuroImage* **17**, 825–841 (2002).

88. Lanczos, C. Evaluation of Noisy Data. *J. Soc. Ind. Appl. Math. Ser. B Numer. Anal.* **1**, 76–85 (1964).

89. Power, J. D. *et al.* Methods to detect, characterize, and remove motion artifact in resting state fMRI. *NeuroImage* **84**, 320–341 (2014).

90. Satterthwaite, T. D. *et al.* An improved framework for confound regression and filtering for control of motion artifact in the preprocessing of resting-state functional connectivity data. *NeuroImage* **64**, 240–256 (2013).

91. Lindquist, M. A., Geuter, S., Wager, T. D. & Caffo, B. S. Modular preprocessing pipelines can reintroduce artifacts into fMRI data. *Hum Brain Mapp* **40**, 2358–2376 (2019).

92. Peirce, J. W. Generating stimuli for neuroscience using PsychoPy. *Front Neuroinform* **2**, 10 (2008).

93. Peirce, J. W. PsychoPy—Psychophysics software in Python. *J Neurosci Meth* **162**, 8–13 (2007).

94. Peirce, J. *et al.* PsychoPy2: Experiments in behavior made easy. *Behav Res Methods* **51**, 195–203 (2019).

95. Levitt, H. Transformed up-down methods in psychoacoustics. *J. Acoust. Soc. Am.* **49**, Suppl 2:467+ (1971).

96. Dumoulin, S. O. & Wandell, B. A. Population receptive field estimates in human visual cortex. *Neuroimage* **39**, 647–660 (2008).

97. Benson, N. C. *et al.* The Human Connectome Project 7 Tesla retinotopy dataset: Description and population receptive field analysis. *J. Vis.* **18**, 23 (2018).

98. Wang, L., Mruczek, R. E. B., Arcaro, M. J. & Kastner, S. Probabilistic Maps of Visual Topography in Human Cortex. *Cereb Cortex New York N Y 1991* **25**, 3911–31 (2014).

99. Destrieux, C., Fischl, B., Dale, A. & Halgren, E. Automatic parcellation of human cortical gyri and sulci using standard anatomical nomenclature. *NeuroImage* **53**, 1–15 (2010).

100. Rossini, P. M. *et al.* Non-invasive electrical and magnetic stimulation of the brain, spinal cord and roots: basic principles and procedures for routine clinical application. Report of an IFCN committee. *Electroen Clin Neuro* **91**, 79–92 (1994).

101. Rossini, P. M. *et al.* Non-invasive electrical and magnetic stimulation of the brain, spinal cord, roots and peripheral nerves: Basic principles and procedures for routine clinical and research application. An updated report from an I.F.C.N. Committee. *Clin Neurophysiol* **126**, 1071–1107 (2015).

102. Stokes, M. G. *et al.* Simple Metric For Scaling Motor Threshold Based on Scalp-Cortex Distance: Application to Studies Using Transcranial Magnetic Stimulation. *J Neurophysiol* **94**, 4520–4527 (2005).

103. Bergen, R. S. van & Jehee, J. F. M. Modeling correlated noise is necessary to decode uncertainty. *Neuroimage* **180**, 78–87 (2018).

104. Bergen, R. S. van & Jehee, J. F. M. TAFKAP: An improved method for probabilistic decoding of cortical activity. *Biorxiv* 2021.03.04.433946 (2021) doi:10.1101/2021.03.04.433946.

105. Yatsenko, D. *et al.* Improved Estimation and Interpretation of Correlations in Neural Circuits. *Plos Comput Biol* **11**, e1004083 (2015).

106. Zhang, R.-Y., Wei, X.-X. & Kay, K. Understanding multivariate brain activity: Evaluating the effect of voxelwise noise correlations on population codes in functional magnetic resonance imaging. *Plos Comput Biol* **16**, e1008153 (2020).

107. Seung, H. S. & Sompolinsky, H. Simple models for reading neuronal population codes. *Proc. Natl. Acad. Sci.* **90**, 10749–10753 (1993).

108. Abbott, L. F. & Dayan, P. The Effect of Correlated Variability on the Accuracy of a Population Code. *Neural Comput.* **11**, 91–101 (1999).

109. Averbeck, B. B., Latham, P. E. & Pouget, A. Neural correlations, population coding and computation. *Nat Rev Neurosci* **7**, 358–366 (2006).

110. Beck, J., Bejjanki, V. R. & Pouget, A. Insights from a Simple Expression for Linear Fisher Information in a Recurrently Connected Population of Spiking Neurons. *Neural Comput.* **23**, 1484–1502 (2011).

111. Kanitscheider, I., Coen-Cagli, R., Kohn, A. & Pouget, A. Measuring Fisher Information Accurately in Correlated Neural Populations. *Plos Comput Biol* **11**, e1004218 (2015).

112. Kanitscheider, I., Coen-Cagli, R. & Pouget, A. Origin of information-limiting noise correlations. *Proc. Natl. Acad. Sci.* **112**, E6973–E6982 (2015).

113. Kriegeskorte, N. & Wei, X.-X. Neural tuning and representational geometry. *Nat. Rev. Neurosci.* **22**, 703–718 (2021).

114. Walther, A. *et al.* Reliability of dissimilarity measures for multi-voxel pattern analysis. *Neuroimage* **137**, 188–200 (2016).

115. Hartlap, J., Simon, P. & Schneider, P. Why your model parameter confidences might be too optimistic. Unbiased estimation of the inverse covariance matrix. *Astron. Astrophys.* **464**, 399–404 (2007).

116. Phipson, B. & Smyth, G. K. Permutation P-values should never be zero: calculating exact P-values when permutations are randomly drawn. *Stat Appl Genet Mol* **9**, Article39 (2010).

117. Koller, M. robustlmm : An R Package for Robust Estimation of Linear Mixed-Effects Models. *J. Stat. Softw.* **75**, (2016).

Supplementary Table 1
Probabilistic Spatial Decoding Results

		Estimation Efficiency		*Posterior SD*	
ROI	*Contrast*	*Δ (deg^{-2})*	*p_{FDR}*	*Δ (°)*	*p_{FDR}*
V1	*Control – CB*	*1.83e-05*	*0.24*	*-2.84*	*0.43*
	Control – IPS	***-4.10e-05***	***0.01***	*2.43*	*0.47*
	Control – sPCS	*-2.27e-05*	*0.17*	*4.30*	*0.19*
V2	*Control – CB*	*2.86e-05*	*0.06*	*-3.88*	*0.17*
	Control – IPS	*-1.01e-05*	*0.50*	*3.12*	*0.21*
	Control – sPCS	*1.40e-05*	*0.34*	*3.09*	*0.18*
V3	*Control – CB*	*-6.00e-06*	*0.70*	*0.96*	*0.78*
	Control – IPS	*-5.21e-06*	*0.70*	***6.32***	***0.03***
	Control – sPCS	***3.89e-05****	***0.01***	*0.49*	*0.86*
V3AB	*Control – CB*	***3.34e-05****	***0.02***	***-9.65****	***0.0004***
	Control – IPS	*2.61e-05*	*0.10*	***-9.37****	***0.0004***
	Control – sPCS	*2.02e-05*	*0.25*	***-10.55****	***0.0004***
IPS0	*Control – CB*	***3.63e-05****	***0.01***	*-4.26*	*0.13*
	Control – IPS	*1.88e-05*	*0.25*	***-11.52****	***0.0004***
	Control – sPCS	*8.99e-06*	*0.52*	*-0.82*	*0.78*
IPS1	*Control – CB*	*-8.33e-07*	*0.94*	***-24.11****	***0.0004***
	Control – IPS	*1.59e-05*	*0.27*	*-5.15*	*0.13*
	Control – sPCS	*1.67e-05*	*0.25*	***-24.70****	***0.0004***
IPS2	*Control – CB*	*-1.26e-05*	*0.40*	***16.86***	***0.0004***
	Control – IPS	*-1.70e-05*	*0.25*	***9.19***	***0.02***
	Control – sPCS	*4.75e-06*	*0.71*	***18.11***	***0.0004***
IPS3	*Control – CB*	*-9.65e-06*	*0.50*	***-9.00****	***0.03***
	Control – IPS	*-1.67e-05*	*0.25*	***-14.35****	***0.0004***
	Control – sPCS	*5.33e-06*	*0.70*	***-19.07****	***0.0004***
sPCS	*Control – CB*	*1.66e-05*	*0.25*	***9.51***	***0.002***
	Control – IPS	*1.86e-05*	*0.25*	***34.45***	***0.0004***
	Control – sPCS	*1.23e-05*	*0.40*	***5.53***	***0.03***
VIIb	*Control – CB*	*1.10e-05*	*0.4463318*	***-40.17****	***0.0005***
	Control – IPS	*3.38e-06*	*0.7737974*	***-42.85****	***0.0005***
	Control – sPCS	*5.60e-06*	*0.6909133*	***-18.10****	***0.0005***

∗ – Deficit relative to control stimulation; Bolded text – $p < 0.05$ corrected

Supplementary Table 2
Spatial Tuning Modulation Results

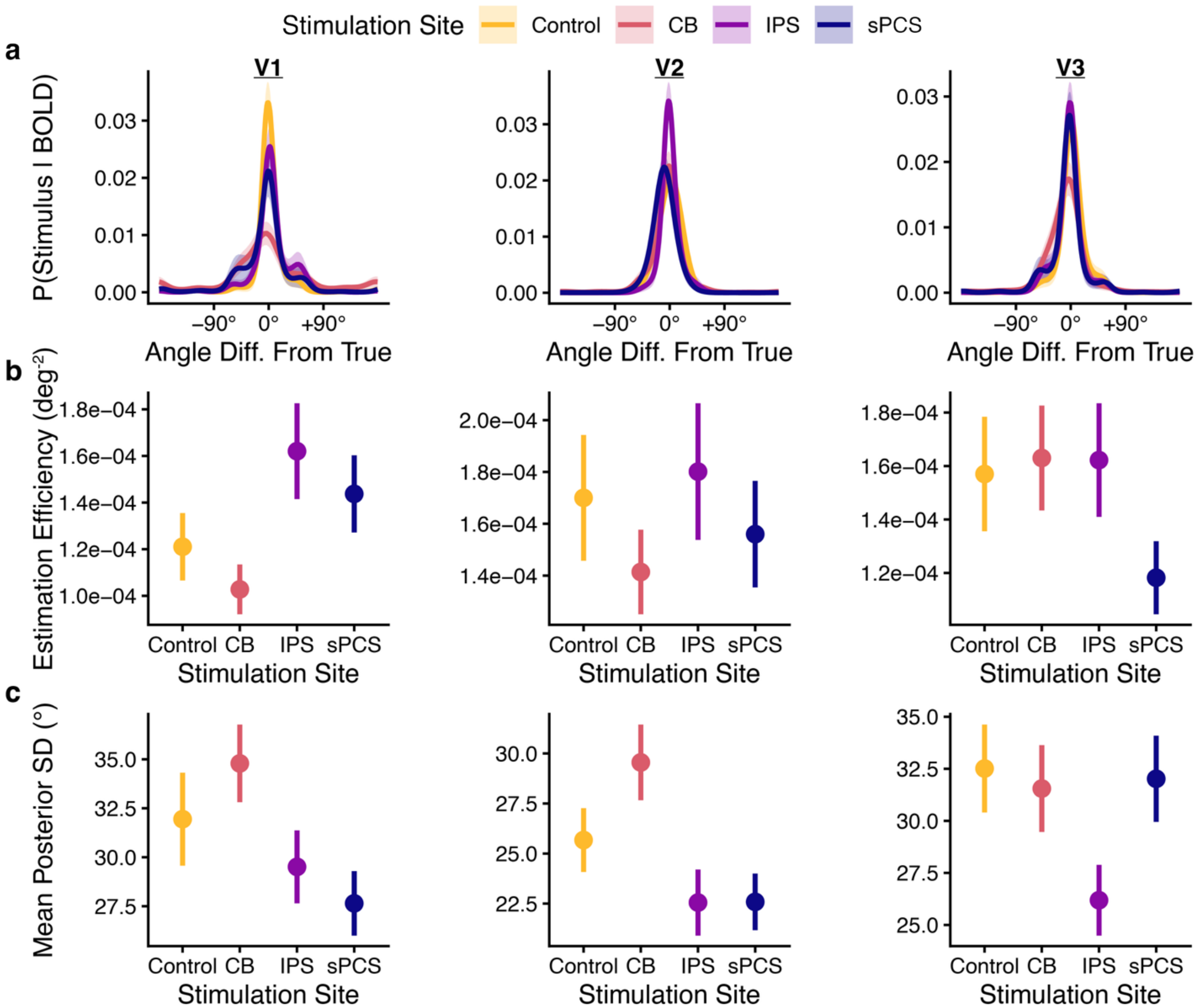
		$\phi_{baseline}$		ϕ_{gain}		ϕ_{size}		ϕ_{var}	
ROI	*Contrast*	Δ	p_{FDR}	Δ	p_{FDR}	Δ	p_{FDR}	Δ	p_{FDR}
V1	*Control – CB*	***0.02****	***0.0002***	***0.24****	***0.0004***	*0.02*	*0.46*	***0.03***	***0.0003***
	Control – IPS	***0.05****	***0.0002***	*0.05*	*0.23*	***0.21***	***0.0008***	*-0.006*	*0.220*
	Control – sPCS	***0.02****	***0.0002***	***-0.22***	***0.001***	*0.01*	*0.67*	***0.03***	***0.0003***
V2	*Control – CB*	***0.01****	***0.0004***	***0.21****	***0.002***	***0.06***	***0.01***	***-0.006****	***0.01***
	Control – IPS	***0.04****	***0.0002***	***-0.19***	***0.01***	***0.13***	***0.0008***	***-0.02****	***0.0003***
	Control – sPCS	***0.02****	***0.0002***	*-0.12*	*0.20*	*0.04*	*0.09*	***-0.01****	***0.0003***
V3	*Control – CB*	***0.01****	***0.0002***	***0.25****	***0.0008***	*0.02*	*0.41*	***0.01***	***0.04***
	Control – IPS	***0.04****	***0.0002***	*-0.08*	*0.413*	*-0.01*	*0.81*	***-0.04****	***0.0003***
	Control – sPCS	***0.03****	***0.0002***	*-0.12*	*0.198*	*0.04*	*0.08*	*-0.006*	*0.08*
V3AB	*Control – CB*	***0.02****	***0.0002***	***0.32****	***0.0004***	*-0.05*	*0.116*	***0.03***	***0.0003***
	Control – IPS	***0.03****	***0.0002***	***0.32****	***0.0004***	*-0.04*	*0.109*	***-0.05****	***0.0003***
	Control – sPCS	***0.03****	***0.0002***	***0.16****	***0.003***	*0.03*	*0.408*	***-0.102****	***0.0003***
IPS0	*Control – CB*	***0.02****	***0.0002***	***0.11****	***0.003***	*0.03*	*0.304*	***-0.04****	***0.005***
	Control – IPS	***0.02****	***0.0002***	***0.21****	***0.0004***	***-0.08****	***0.0007***	*-0.01*	*0.541*
	Control – sPCS	***0.03****	***0.0002***	*-0.04*	*0.35*	***0.06***	***0.0388***	***-0.07****	***0.0003***
IPS1	*Control – CB*	***0.02****	***0.0002***	***0.29****	***0.0005***	*-0.01*	*0.78*	***-0.105****	***0.0003***
	Control – IPS	***0.02****	***0.0002***	***0.09****	***0.04***	***0.07****	***0.0008***	***0.0860***	***0.0003***
	Control – sPCS	***0.03****	***0.0002***	***0.27****	***0.0004***	*0.02*	*0.45*	***0.102***	***0.0003***
IPS2	*Control – CB*	***0.04****	***0.0002***	***0.12****	***0.0004***	***0.151***	***0.0008***	***0.03***	***0.02***
	Control – IPS	***0.03****	***0.0002***	***0.13****	***0.0004***	*0.0251*	*0.180*	***0.09***	***0.0003***
	Control – sPCS	***0.03****	***0.0002***	*0.02*	*0.45*	***0.168***	***0.0008***	***0.04***	***0.002***
IPS3	*Control – CB*	***0.03****	***0.0002***	***0.24****	***0.0004***	***-0.08****	***0.03***	***-0.09****	***0.0003***
	Control – IPS	***0.03****	***0.0002***	***0.22****	***0.0004***	*0.02*	*0.65*	*-0.001*	*0.97*
	Control – sPCS	***0.02****	***0.0002***	***0.23****	***0.0004***	***-0.07****	***0.02***	***-0.05****	***0.002***
sPCS	*Control – CB*	***0.002****	***0.003***	*-0.005*	*0.61*	*0.006*	*0.554*	***-0.125****	***0.0003***
	Control – IPS	***0.004****	***0.0002***	*-0.03*	*0.08*	***0.05***	***0.003***	***0.133***	***0.0003***
	Control – sPCS	*-0.001*	*0.09*	*-0.008*	*0.44*	***-0.02****	***0.02***	***-0.289****	***0.0003***
VIIb	*Control – CB*	***0.01****	***0.0002***	***0.32****	***0.0004***	***-0.14****	***0.0008***	***-0.07****	***0.009***
	Control – IPS	***0.03****	***0.0002***	***0.46****	***0.0004***	***-0.13****	***0.0008***	***-0.21****	***0.0003***
	Control – sPCS	***0.01****	***0.0002***	***0.257****	***0.0004***	***-0.08****	***0.03***	*-0.001*	*0.97*

∗ – Deficit relative to control stimulation; Bolded text – $p < 0.05$ corrected

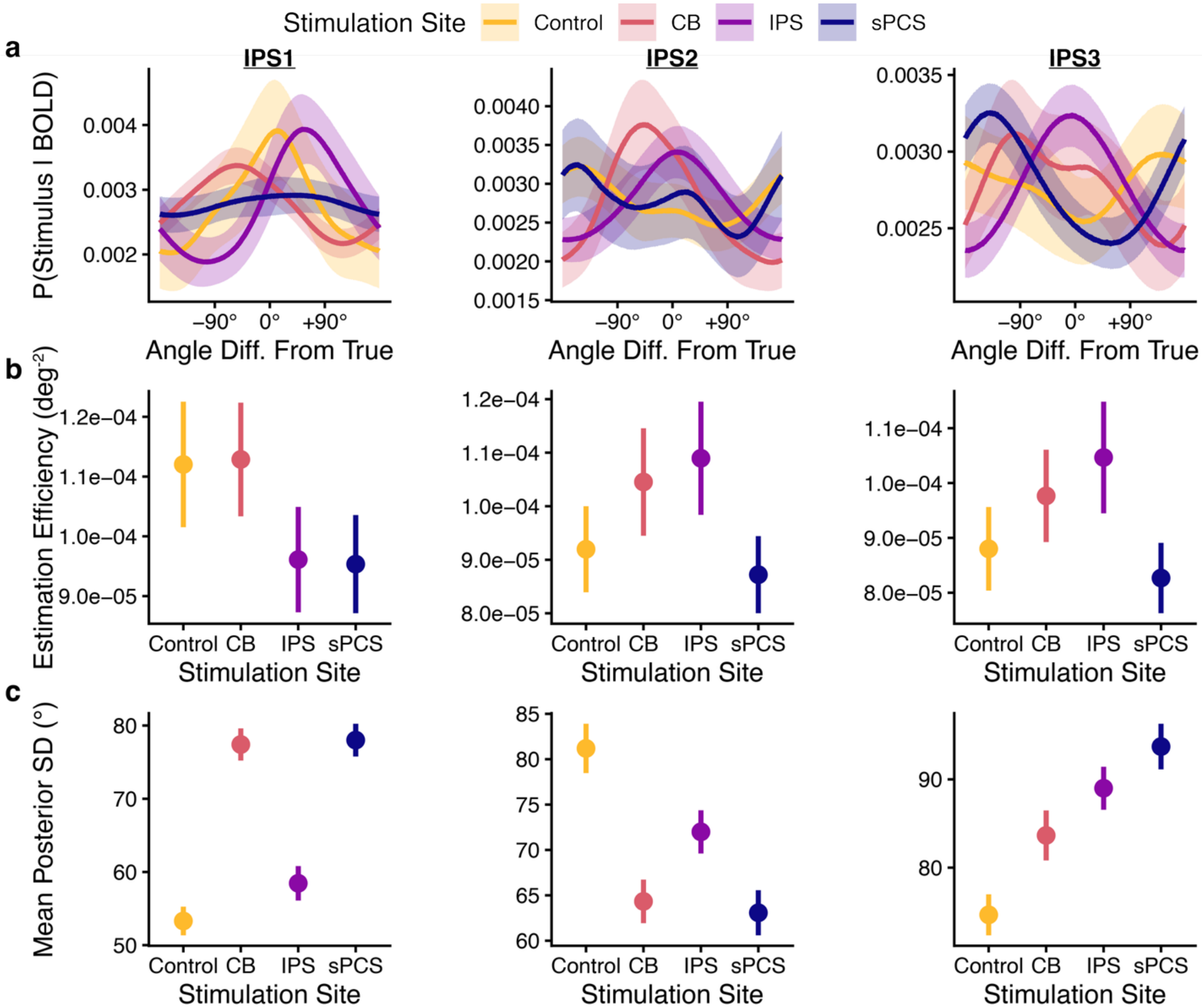
Supplementary Table 3
Cross-Validated Fisher Information Results

ROI	Contrast	$\bar{I}_{cv}$ Δ (rad^{-2})	$\bar{I}_{cv}$ p_{FDR}
V1	*Control – CB*	***2.91****	***0.006***
	Control – IPS	*-10.1*	*0.50*
	Control – sPCS	*0.407*	*0.78*
V2	*Control – CB*	*-1.05*	*0.12*
	Control – IPS	***-2.36***	***0.004***
	Control – sPCS	*-0.600*	*0.38*
V3	*Control – CB*	*0.545*	*0.44*
	Control – IPS	*-1.92*	*0.21*
	Control – sPCS	*-1.85*	*0.08*
V3AB	*Control – CB*	***1.63****	***0.002***
	Control – IPS	***1.58****	***0.002***
	Control – sPCS	***1.73****	***0.002***
IPS0	*Control – CB*	*0.552*	*0.08*
	Control – IPS	***0.675****	***0.01***
	Control – sPCS	*0.459*	*0.21*
IPS0	*Control – CB*	***0.0928****	***0.003***
	Control – IPS	***0.0618****	***0.01***
	Control – sPCS	***0.10****	***0.002***
IPS1	*Control – CB*	*-0.003*	*0.995*
	Control – IPS	*-0.0008*	*0.995*
	Control – sPCS	*0.0003*	*0.995*
IPS2	*Control – CB*	*0.008*	*0.45*
	Control – IPS	***0.02****	***0.003***
	Control – sPCS	***0.02****	***0.003***
IPS3	*Control – CB*	*0.55*	*0.08*
	Control – IPS	***0.68****	***0.01***
	Control – sPCS	*0.459*	*0.21*
sPCS	*Control – CB*	*-0.0004*	*0.177*
	Control – IPS	***-0.006***	***0.003***
	Control – sPCS	*-0.0003*	*0.425*
Lobule VIIb	*Control – CB*	***0.105****	***0.002***
	Control – IPS	***0.105****	***0.002***
	Control – sPCS	***0.0888****	***0.002***

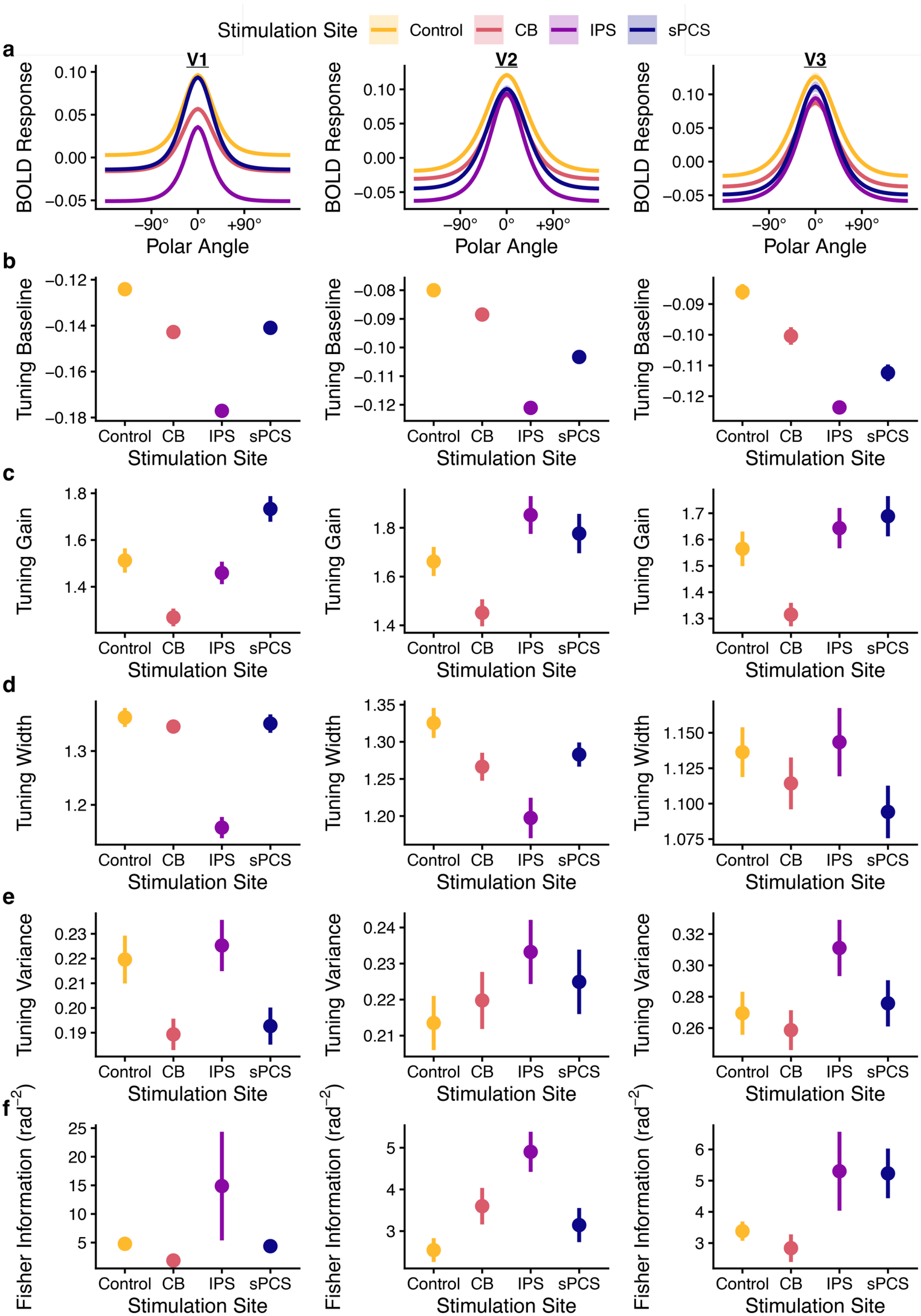
∗ – Deficit relative to control stimulation; Bolded text – $p < 0.05$ corrected



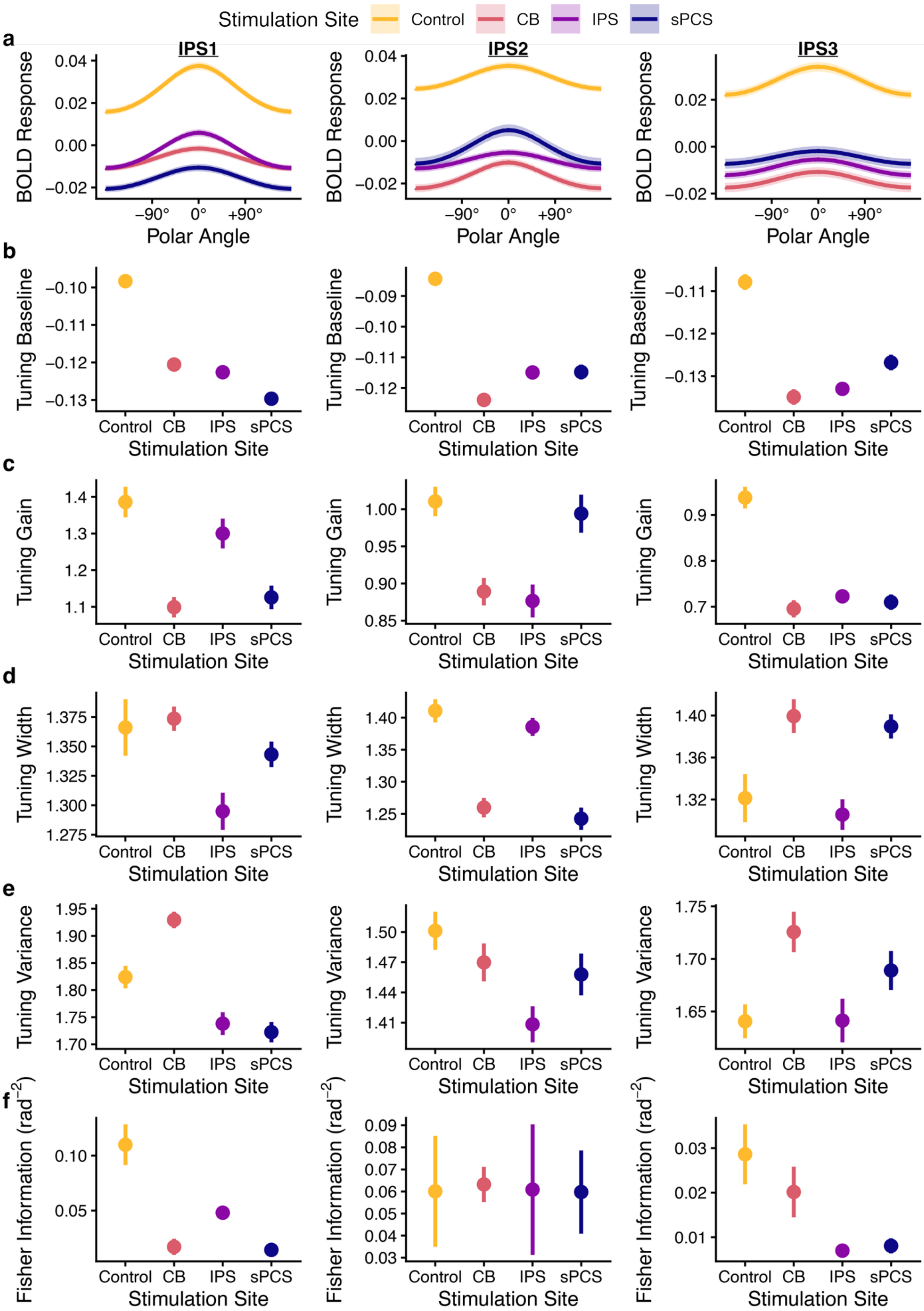
Supplementary Figure 1. Visual cortex spatial decoding. (a) Mean posterior probability distribution. Averaging trial-aligned posterior distributions produces a session-level probability density over stimulus locations for each region of interest (ROI). Sharper peaks around 0° reflect better decoding of the remembered spatial location. (b) Estimation efficiency (inverse of mean circular squared error between the posterior mean and the true stimulus) reflecting decoding accuracy for each session and region. (c) Mean posterior standard deviation reflecting decoding uncertainty. Error bars reflect a bootstrap estimate of standard error. Each column reflects results for one ROI.



Supplementary Figure 2. Parietal cortex spatial decoding. (a) Mean posterior probability distribution. Averaging trial-aligned posterior distributions produces a session-level probability density over stimulus locations for each region of interest (ROI). Sharper peaks around 0° reflect better decoding of the remembered spatial location. (b) Estimation efficiency (inverse of mean circular squared error between the posterior mean and the true stimulus) reflecting decoding accuracy for each session and region. (c) Mean posterior standard deviation reflecting decoding uncertainty. Error bars reflect a bootstrap estimate of standard error. Each column reflects results for one ROI.


Stimulation Site
Control
CB
IPS
sPCS
a
V1
V2
V3
BOLD Response
Polar Angle
−90°
0°
+90°
b
Tuning Baseline
c
Tuning Gain
d
Tuning Width
e
Tuning Variance
f
Fisher Information (rad^{-2})
Stimulation Site

Supplementary Figure 3. Visual cortex spatial tuning modulation during WM maintenance. (a) Mean tuning curve across voxels for each stimulation session within each ROI. For visualization, the tuning curves are demeaned across sessions. Shaded ribbons reflect bootstrap estimate of SEM across voxels. (b) Spatial tuning curve baseline for each session and ROI. This value reflects stimulus-independent BOLD activity during the delay-period. (c) Spatial tuning curve gain for each session and ROI. This value reflects the BOLD response for preferred stimulus locations relative to non-preferred locations. (d) Spatial tuning curve width for each session and ROI. (e) Spatial tuning curve variance for each session and ROI reflecting the distribution of BOLD responses for repeated presentations of the same stimulus. (f) Cross-validated Fisher information for each session and ROI. This value reflects how sensitively a region's mean response varies with spatial location relative to its noise covariance, thus characterizing the intrinsic representational capacity supported by the underlying population code. This measure represents the net impact of all TMS-induced changes in spatial tuning. Error bars across b-f reflect a bootstrap estimate of SEM.


Stimulation Site
Control
CB
IPS
sPCS
a
IPS1
IPS2
IPS3
BOLD Response
Polar Angle
b
Tuning Baseline
c
Tuning Gain
d
Tuning Width
e
Tuning Variance
f
Fisher Information (rad⁻²)
Control CB IPS sPCS
Stimulation Site

Supplementary Figure 3. Parietal cortex spatial tuning modulation during WM maintenance. (a) Mean tuning curve across voxels for each stimulation session within each ROI. For visualization, the tuning curves are demeaned across sessions. Shaded ribbons reflect bootstrap estimate of SEM across voxels. (b) Spatial tuning curve baseline for each session and ROI. This value reflects stimulus-independent BOLD activity during the delay-period. (c) Spatial tuning curve gain for each session and ROI. This value reflects the BOLD response for preferred stimulus locations relative to non-preferred locations. (d) Spatial tuning curve width for each session and ROI. (e) Spatial tuning curve variance for each session and ROI reflecting the distribution of BOLD responses for repeated presentations of the same stimulus. (f) Cross-validated Fisher information for each session and ROI. This value reflects how sensitively a region's mean response varies with spatial location relative to its noise covariance, thus characterizing the intrinsic representational capacity supported by the underlying population code. This measure represents the net impact of all TMS-induced changes in spatial tuning. Error bars across b-f reflect a bootstrap estimate of SEM.